

Bacteriological Examination of Water: Qualitative Tests

C. Results of Completed Test

Record the results of completed tests for each water sample that was positive on the confirmed test.

WATER SAMPLE (SOURCE)	LACTOSE FERMENTATION RESULTS	MORPHOLOGY	EVALUATION

D. Questions

- Does a positive presumptive test mean that the water is absolutely unsafe to drink? _____
Explain: _____
- What might cause a false positive presumptive test? _____

- List three characteristics required of a good sewage indicator: a. _____
b. _____ c. _____
- What enteric **bacterial** diseases are transmitted in polluted water? _____

- Name one or more **protozoan** diseases transmitted by polluted water. _____

- Why don't health departments routinely test for pathogens instead of using a sewage indicator?

- Give the functions of the various media used in these tests:
Lactose broth: _____
Levine EMB agar: _____

Nutrient agar slant: _____

- What media, other than the ones used here, can be used for confirmatory tests? _____

LABORATORY REPORT

64, 65

Student: _____

Desk No.: _____ Section: _____

Ex. 64 Membrane Filter Method

A. Tabulation

A table similar to the one below will be provided for you, either on the chalkboard or as a photocopy. Record your coliform count on it. Once all data are available, complete this table.

SAMPLE	SOURCE	COLIFORM COUNT	AMOUNT OF WATER FILTERED	MPN*
A				
B				
C				
D				
E				
F				
G				
H				

*MPN = $\frac{\text{Coliform Count} \times 100}{\text{Amount of Water Filtered}}$

B. Questions

1. Give two limitations of the membrane filter technique:

2. Even if the membrane filter removed all bacteria from water being tested, is the water that passes through sterile? _____ Explain: _____

3. List some other applications of membrane filter technology. _____

Standard Plate Count: A Quantitative Test

**Ex. 65 Standard Plate Count:
A Quantitative Test****A. Tabulation of Results**

After you have made your plate counts, record your results on the chalkboard and on the following table. After all students have recorded their results, complete this table.

SAMPLE	SOURCE	PLATE COUNT (AVERAGE)	DILUTION	ORGANISMS PER ML
A				
B				
C				
D				
E				
F				
G				
H				
I				

What generalizations, if any, can be drawn from these results?

B. Questions

1. What kinds of organisms in water will not produce colonies on TGEA?

2. Would a high plate count indicate that the water is unsafe to drink? _____

Explain: _____

LABORATORY REPORT

66, 67

Student: _____

Desk No.: _____ Section: _____

Ex. 66 Standard Plate Count of Milk

A. Tabulation of Results

After you have made your plate count, record your results on the following table. Get the results of the other milk sample from some other member of the class.

TYPE OF MILK	PLATE COUNT	DILUTION	ORGANISMS PER ML
High-quality			
Poor-quality			

B. Questions

- Do plate count figures represent numbers of organisms or numbers of clumps of bacteria?

- What are some factors that will produce errors in the SPC technique?

- What might be the explanation of a very high count in raw milk that has been properly refrigerated from the time of collection?

- What is the most common source of bacteria in milk? _____
- Why is milk a more suitable vector of disease than water? _____

- What infectious diseases of cows can be transmitted to humans via milk? _____

Direct Microscopic Count of Organisms in Milk: The Breed Count

Ex. 67 Direct Microscopic Count of Organisms in Milk: The Breed Count

A. Microscope Factor

Show your computations in determining the MF of your microscope.

1. Field diameter: _____ millimeters
2. Field area (πr^2): _____ square millimeters
_____ square centimeters
3. Number of fields in one square centimeter: _____
4. Microscope factor ($100 \times$ no. of fields): _____

B. Cell Counts

Record the counts of each field, total them, and determine their average counts. Get the results of the other type of milk from a class member.

	GOOD MILK			POOR MILK		
	Bacterial Cells	Bacterial Clumps	Leukocytes	Bacterial Cells	Bacterial Clumps	Leukocytes
1						
2						
3						
4						
5						
Total						
Av.						

Determine the number of cells, clumps, and leukocytes per ml for each type of milk.

Good Milk

Bacterial Cells: _____

Bacterial Clumps: _____

Leukocytes: _____

Poor Milk

Bacterial Cells: _____

Bacterial Clumps: _____

Leukocytes: _____

C. Questions

1. Should milk from a healthy cow be completely free of bacteria? _____
2. Are leukocytes in milk always an indication of infection (mastitis)? _____
3. Why are direct counts higher than standard plate counts? _____

LABORATORY REPORT

68

Student: _____

Desk No.: _____ Section: _____

Reductase Test

1. How would you grade the two samples of milk that you tested? Give the MBRT for each one.

Sample A: _____

Sample B: _____

2. Is milk with a short reduction time necessarily unsafe to drink? _____

Explain: _____

3. What other dye can be substituted for methylene blue in this test? _____

4. What advantage do you see in this method over the direct count method? _____

5. What kinds of organisms may be plentiful in a milk sample, yet give a negative reductase test?

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LABORATORY REPORT

69

Student: _____

Desk No.: _____ Section: _____

Bacterial Counts of Foods

A. Tabulation of Results

Record your count and the bacterial counts of various other foods made by other students.

TYPE OF FOOD	PLATE COUNT	DILUTION	ORGANISMS PER ML

B. Questions

- Why is there such great variability in organisms per ml between different kinds of food? _____

- What dangers and undesirable results may occur from ground meats of high bacterial counts?

- What bacterial pathogens might be present in frozen foods? _____

- What harm can result from repeated thawing and freezing of foods? _____

Bacterial Counts of Foods

5. What precautions are taken to prevent the spoilage of foods? _____

6. Which methods in question 5 are most effective? _____

Least effective? _____

LABORATORY REPORT

70, 71

Student: _____

Desk No.: _____ Section: _____

Ex. 70 Microbial Spoilage of Canned Food

A. Results

Record your observations of the effects of each organism on the cans of vegetables. Share results with other students.

ORGANISM	PEAS		CORN	
	Gas Production + or -	Odor	Gas Production + or -	Odor
<i>E. coli</i>				
<i>B. coagulans</i>				
<i>B. stearothermophilus</i>				
<i>C. sporogenes</i>				
<i>C. thermosaccharolyticum</i>				

B. Microscopy

After making gram-stained and spore-stained slides of all organisms from the canned food extracts, sketch in representatives of each species:

--	--	--	--	--

*E. coli**B. coagulans**B. stearothermophilus**C. sporogenes**C. thermosac-*
*charolyticum***C. Questions**

1. Which organisms, if any, caused “flat sour spoilage”? _____

2. Which organisms, if any, caused “T.A. spoilage”? _____

3. Which organisms, if any, caused “stinker spoilage”? _____

4. Does “flat sour” cause a health problem? _____
5. Describe how typical spoilage resulting in botulism occurs. _____

Microbial Spoilage of Refrigerated Meat

Ex. 71 Microbial Spoilage of Refrigerated Meat

A. Results

1. After performing the colony count on your best plates, how many psychrophilic-psychrotrophic bacteria do you find were present in each gram of the meat sample? _____
2. Describe the appearance of the organism that you stained with Gram's stain. _____

B. Questions

1. Differentiate between the following:
Psychrophile: _____

Psychrotroph: _____

2. Why was it necessary to incubate the plates for 2 weeks? _____

3. List some genera of bacteria that might be psychrophilic or psychrotrophic in the meat sample:
Gram-negative types: _____

Gram-positive types: _____

4. List three or four pathogenic psychrotrophs that might be found in refrigerated meat: _____

5. What are the probable sources of psychrophilic microbes found growing in meat? _____

6. What types of measures can be taken to prevent spoilage of meats by psychrophilic-psychrotrophic bacteria? _____

LABORATORY REPORT

72, 73

Student: _____

Desk No.: _____ Section: _____

Ex. 72 Microbiology of Fermented Beverages

A. Results

Record here your observations of the fermented product:

Aroma: _____

pH: _____

H₂S production: _____

B. Questions

1. Why must the fermentor be sealed? _____

Why with a balloon? _____

2. What compound in the grape juice is being fermented? _____

3. Why would production of hydrogen sulfide by the yeast be of importance? _____

4. Why are we concerned about the pH of the fruit juice and the wine? _____

5. What happens to wine if *Acetobacter* takes over? _____

6. What process can be used to prevent the action of *Acetobacter* in the production of wine and beer? _____

Ex. 73 Microbiology of Fermented Milk Products

A. Results

1. Record here your observations of the fermented product:

Color: _____

Aroma: _____

Texture: _____

Taste: _____

Microbiology of Fermented Milk Products

2. In the space provided below sketch in the microscopic appearance of the organisms as seen on the slide stained with methylene blue.

B. Questions

1. What type of fermentation is involved in the production of yogurt? _____

2. What is the nutritional value of yogurt? _____

LABORATORY REPORT

74, 75

Student: _____

Desk No.: _____ Section: _____

Ex. 74 Mutant Isolation by Gradient Plate Method

A. Results

1. How many colonies of *E. coli* did you count in the high concentration area of the plate?

2. When the streptomycin-resistant colonies were smeared with a loop toward the high concentration side and reincubated, did they continue to grow in the new area?

What does this result indicate? _____

Ex. 75 Mutant Isolation by Replica Plating

A. Tabulation of Results

Count the colonies that occur on both plates and record the information on the chalkboard on a table similar to the one below. After all students have recorded their counts, complete this table.

STUDENT INITIALS	NUMBER OF COLONIES		STUDENT INITIALS	NUMBER OF COLONIES	
	Nutrient Agar	Streptomycin Agar		Nutrient Agar	Streptomycin Agar
TOTALS			TOTALS		
AVERAGE PER PLATE			AVERAGE PER PLATE		

Mutant Isolation by Replica Plating

B. Questions

1. After determining the average number of streptomycin-resistant colonies per plate, calculate the mutability rate of the organisms, assuming that there were 100,000,000 organisms per milliliter in the original culture. Show your mathematical computations.

Mutability rate: _____

2. What does replica plating of mutants attempt to prove that is not established by the gradient plate method?

3. How can the frequency of mutation in bacteria be increased?

4. In what way does the presence of an antibiotic increase the numbers of resistant forms of bacteria?

LABORATORY REPORT

76

Student: _____

Desk No.: _____ Section: _____

Bacterial Mutagenicity and Carcinogenesis: The Ames Test

A. Tabulation of Results

Record the results of your tests in the following table and on a similar table on the chalkboard. Also record the results of substances tested by other students. A positive result will exhibit a zone of colonies similar to the zone shown on the plate in illustration 5, figure 76.1.

TEST SUBSTANCE	RESULT (+ or -)	TEST SUBSTANCE	RESULT (+ or -)	TEST SUBSTANCE	RESULT (+ or -)

B. Questions

- Did you observe a zone of inhibition between the growing colonies and the impregnated disk on your positive plates? _____
What is the cause of such a zone? _____
- Differentiate between the following:
Prototroph: _____
Auxotroph: _____
- Define **back mutation**: _____
- List two characteristics of the Ames test that made this test so much superior to previous mutagenesis tests: _____

Bacterial Mutagenicity and Carcinogenesis: The Ames Test

5. Does the fact that a chemical substance is carcinogenic in animals necessarily mean that it is also carcinogenic in humans? _____

LABORATORY REPORT

77

Student: _____

Desk No.: _____ Section: _____

A Synthetic Epidemic

A. Tabulation of Results

Record in the table below the information that has been tabulated on the chalkboard. The SHAKER is the person designated by the instructor to shake hands with another class member. The SHAKEE is the individual chosen by the shaker. For **Procedure A** a blue color is positive; yellow or brown is negative. For **Procedure B** red colonies (*S. marcescens*) is positive; no red colonies is negative.

SHAKER Round 1	RESULT + or -	SHAKEE Round 1	RESULT + or -	SHAKER Round 2	RESULT + or -	SHAKEE Round 2	RESULT + or -
1.				1.			
2.				2.			
3.				3.			
4.				4.			
5.				5.			
6.				6.			
7.				7.			
8.				8.			
9.				9.			
10.				10.			
11.				11.			
12.				12.			
13.				13.			
14.				14.			
15.				15.			
16.				16.			
17.				17.			
18.				18.			
19.				19.			
20.				20.			
21.				21.			
22.				22.			
23.				23.			
24.				24.			

A Synthetic Epidemic

B. Questions

1. Who in the group was “patient zero,” the starter of the epidemic? _____
2. How many carriers resulted after
Round 1? _____ Round 2? _____
3. If this were a real infectious agent, such as a cold virus or influenza, list some other factors in transmission besides the ones we tested:

4. How would it have been possible to stop this infection cycle? _____

Student: _____

78

Desk No.: _____ **Section:** _____

The Staphylococci: Isolation and Identification

A. Tabulation

At the beginning of the third laboratory period, the instructor will construct a chart similar to this one on the chalkboard. After examining your mannitol salt agar and staphylococcus medium 110 plates, record the presence (+) or absence (–) of staphylococcus growth in the appropriate columns. After performing coagulase tests on the various isolates record the results also as (+) or (–) in the appropriate columns.

[illegible]

The Staphylococci: Isolation and Identification**B. Microscopy**

Provide drawings here of the various isolates as seen under oil immersion (gram staining).

UNKNOWN-CONTROL	NOSE	FOMITE
-----------------	------	--------

C. Percentages

From the data in the table on the previous page, determine the incidence (percentage) of individuals and fomites that harbor coagulase-positive and coagulase-negative staphylococci in this experiment.

SOURCE	TOTAL TESTED	TOTAL POSITIVE	PERCENTAGE POSITIVE	TOTAL NEGATIVE	PERCENTAGE NEGATIVE
Humans (Nose)					
Fomites					

D. Record of Test Results

Record here the results of each test performed in this experiment. Under GRAM STAIN indicate cellular arrangement as well as Gram reaction.

ISOLATE	GRAM STAIN	ALPHA TOXIN	MANNITOL (ACID)	COAGULASE
Unknown-Control No. ____				
Nose Isolate No. 1				
Nose Isolate No. 2				
Fomite Isolate				

E. Final Determination

Record here the name of your unknown-control. If API Staph-Ident miniaturized multitest strips are available, confirm your conclusions by testing each isolate. See Exercise 55.

Name of unknown-control: _____

Staph-Ident results: _____

F. Questions

1. What are nosocomial infections? _____

2. What bacterial organism causes most nosocomial infections? _____

Student: _____

79

Desk No.: _____ **Section:** _____

The Streptococci: Isolation and Identification

A. Tabulation of Pharynx Isolates

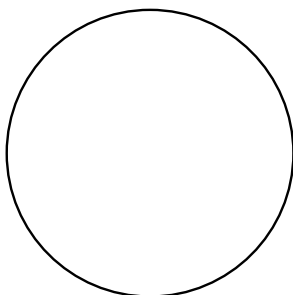
The instructor will construct a chart similar to this one on the chalkboard. After examining the blood agar plates that were inoculated with pharynx organisms, record the types and size range of colonies that are present on your plates. Record these data first on this table, then on the chalkboard. After all students have recorded their results on the board, complete the tabulation of their results here, also. The names of the organisms will not be recorded until all tests are completed.

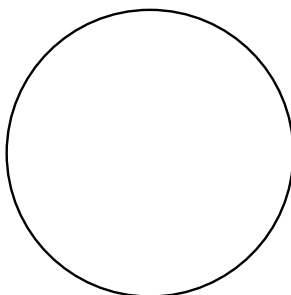
[illegible]

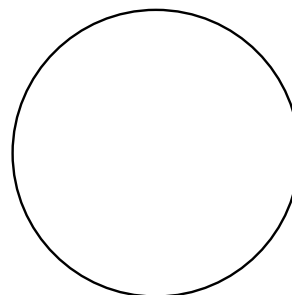
The Streptococci: Isolation and Identification

B. Microscopy

Provide drawings here of the various pharyngeal isolates as seen under oil immersion (gram staining).







C. Percentages

From the data in the table on the previous page, calculate the percentages for each type of streptococci that were isolated from classmates.

S. pyogenes: _____

Group C streptococci: _____

S. agalactiae: _____

Group D enterococci: _____

S. pneumoniae: _____

Group D nonenterococci: _____

Viridans streptococci: _____

D. Record of Test Results

Record here all information pertaining to the identification of pharyngeal isolates and unknowns.

SOURCE OF UNKNOWN	Hemolysis	Bacitracin Susceptibility	CAMP Reaction	Hippurate Hydrolysis	SXT Sensitivity	Bile Esculin Hydrolysis	Tolerance to 6.5% NaCl	Susceptibility to Optochin	Bile Solubility

E. Final Determination

Record here the identities of your various isolates and unknowns:

Pharyngeal isolates: _____

Unknowns: _____

The Streptococci: Isolation and Identification

F. Questions

Record the answers for the following questions in the answer column.

1. What two physiological tests are significant in the identification of *S. agalactiae*?
2. If an alpha hemolytic streptococcus is able to hydrolyze bile esculin, what test can be used to tell whether the organism is an enterococcus?
3. What test is used for differentiating group A from group C streptococci if both organisms are bacitracin-susceptible?
4. What two tests are used to differentiate pneumococci from the viridians group?
5. What test is used for differentiating *S. pyogenes* from other beta hemolytic streptococci?
6. Hemolysis in streptococci can only be evaluated when the colonies develop _____ (aerobically or anaerobically) in blood agar.
7. Which streptococcal species is frequently present in the vagina of third-trimester pregnant women?
8. Only one beta hemolytic streptococcus is primarily of human origin. Which one is it?
9. Who developed the system of classifying streptococci into groups A, B, C, etc.?
10. Who is credited with grouping streptococci according to the type of hemolysis?
11. Which streptococcal species is seen primarily as paired cells (diplococci)?
12. Name two species of streptococci that are implicated in dental caries.
13. Where in the body can *S. bovis* be found?
14. After performing all physiological tests, what type of tests must be performed to confirm identification?
15. Which hemolysin produced by *S. pyogenes* is responsible for the beta-type hemolysis that is characteristic of this organism?

Answers

- 1.a. _____
b. _____
2. _____
3. _____
- 4.a. _____
b. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
- 12.a. _____
b. _____
13. _____
14. _____
15. _____

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LABORATORY REPORT

80

Student: _____

Desk No.: _____ Section: _____

Gram-Negative Intestinal Pathogens

A. Unknown Identification

1. What was the genus of your unknown?

_____	_____
Genus	No.

2. What problems, if any, did you encounter?

3. Now that you know the genus of your unknown, what steps would you follow to determine the species?

B. General Questions

1. Why are bile salts and sodium desoxycholate used in certain selective media in this exercise?

2. How can one identify coliforms on MacConkey agar? _____

3. How does one differentiate *Salmonella* from *Shigella* colonies on XLD agar? _____

4. What characteristics do the salmonella and shigella have in common?

Gram-Negative Intestinal Pathogens

5. How do the salmonella and shigella differ?

6. What restrictions might be placed on a person who is a typhoid carrier?

LABORATORY REPORT

81

Student: _____

Desk No.: _____ Section: _____

Urinary Tract Pathogens

A. Quantitative Evaluation

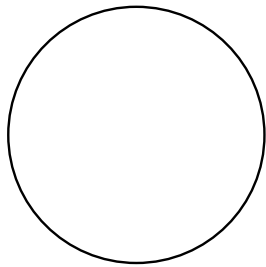
After counting the colonies on the TSA plate, record the count as follows:

Number of colonies: _____

Dilution: _____

No. of organisms/ml of urine: _____

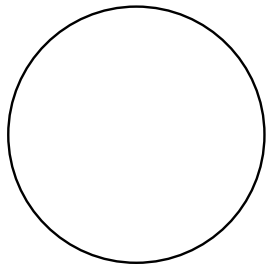
Gram-Stained Slide. If organisms are seen on a gram-stained slide of an uncentrifuged sample, sketch in color in the circle below.



Conclusion: Do the plate count and gram-stained slide of the uncentrifuged sample provide presumptive evidence of a urinary infection?

B. Microscopic Study (Centrifuged Sample)

Illustrate in the circle below the microscopic appearance of a centrifuged sample.



Conclusion: Describe here the morphological appearance of the predominant organism seen:

C. Culture Analyses

After studying the organisms on the three plates and thioglycollate medium, what organism do you believe is causing the infection?

Organism: _____

What further testing should be performed for confirmation?

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LABORATORY REPORT

82, 83

Student: _____

Desk No.: _____ Section: _____

Ex. 82 Slide Agglutination Test: Serological Typing

1. Record the unknown number that proved to be a salmonella. _____
2. Why was phenolized saline used instead of plain physiological saline? _____

3. If your results were negative for both cultures, what might be the explanation? _____

Ex. 83 Slide Agglutination (Latex) Test: For *S. aureus* Identification

1. If your test turned out to be positive for *S. aureus*, record the degree of positivity here: _____
2. What other test for *S. aureus* is highly correlated with this test? _____
3. What two kinds of antibodies are attached to the latex particles in the Difco latex reagent?
a. _____ b. _____
4. What role, if any, do the staphylococci cells play in this reaction? _____

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LABORATORY REPORT

84, 85

Student: _____

Desk No.: _____ Section: _____

Ex. 84 Tube Agglutination Test: The Heterophile Antibody Test

1. What was the titer of the serum that you tested? _____
2. For what disease is this diagnostic test used? _____
3. Below what titer would this test be considered to be negative? _____
4. What is the name of the virus that causes this disease? _____
5. What is unusual about a heterophile antibody? _____

6. *If you are not going to perform the Widal test (Ex. 85), answer all the questions below except for question 1. It may be necessary for you to read the introduction to Ex. 85 for some of the answers.*

Ex. 85 Tube Agglutination Test: The Widal Test

1. What was the titer of the serum that you tested? _____
2. What is the exact meaning of the "titer" of a serum? _____

3. Differentiate between the following:
Serum: _____

Antiserum: _____

Antitoxin: _____

4. How would you prepare antiserum for an organism such as *E. coli*? _____

Tube Agglutination Test: The Widal Test

5. Indicate the type of antigen (*soluble protein*, *red blood cells*, or *bacteria*) that is used for each of the following serological tests:

Agglutination: _____

Precipitation: _____

Hemolysis: _____

6. For which one of the above tests is complement necessary? _____

LABORATORY REPORT

86

Student: _____

Desk No.: _____ Section: _____

Phage Typing

1. To which phage types was this strain of *S. aureus* susceptible?

2. To what lytic group does this strain of staphylococcus belong?

3. In what way can bacteriophage alter the genetic structure of a bacterium?

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LABORATORY REPORT

87–89

Student: _____

Desk No.: _____ Section: _____

Ex. 87 White Blood Cell Study: The Differential WBC Count

A. Tabulation of Results

As you move the slide in the pattern indicated in figure 87.4, record all the different types of cells in the following table. Refer to figures 87.1 and 87.2 for cell identification. Use this method of tabulation: ~~||||~~ ~~||||~~
11. Identify and tabulate 100 leukocytes. Divide the total of each kind of cell by 100 to determine percentages.

NEUTROPHILS	LYMPHOCYTES	MONOCYTES	EOSINOPHILS	BASOPHILS
Total				
Percent				

B. Questions

- Were your percentages for each type within the normal ranges? _____

- What errors might one be likely to make when doing this count for the first time? _____

- Differentiate between the following:
Cellular immunity: _____
Humoral immunity: _____
- Do cellular and humoral immunity work independently? _____ Explain: _____

Total WBC Count and Blood Grouping

Ex. 88 Total WBC Count

A. Calculations

Using the formula provided on page 295, calculate the number of leukocytes per cubic millimeter.

Total Count: _____

B. Questions

1. What is the normal WBC count range? _____
2. List several causes of a high WBC count: _____

Ex. 89 Blood Grouping

1. On the basis of this test, what is your blood type? _____
2. What antibodies are present in each type of blood?
Type A: _____ Type B: _____ Type AB: _____ Type O: _____
3. Why does a person of type A blood go into a transfusion reaction when given type B blood? _____

4. Why can Rh-positive blood be given only once to a person who is Rh-negative? _____

LABORATORY REPORT

90

Student: _____

Desk No.: _____ Section: _____

The Snyder Caries Susceptibility Test

1. What degree of caries susceptibility was indicated for you as a result of this test? _____

2. Is this substantiated by the amount of dental work you have had or should have had on your teeth?

3. What factors could affect the reliability of this test? _____

4. To increase the validity of this test how many times should it be performed and at what time of the day?

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Descriptive Chart

STUDENT: _____		Habitat: _____ Culture No.: _____		
LAB SECTION: _____		Source: _____		
		Organism: _____		
MORPHOLOGICAL CHARACTERISTICS		PHYSIOLOGICAL CHARACTERISTICS		
Cell Shape: Arrangement: Size: Spores: Gram's Stain: Motility: Capsules: Special Stains:	TESTS		RESULTS	
	Fermentation	Glucose		
		Lactose		
		Sucrose		
		Mannitol		
	Hydrolysis	Gelatin Liquefaction		
		Starch		
		Casein		
		Fat		
	IMViC	Indole		
		Methyl Red		
		V-P (acetylmethylcarbinol)		
		Citrate Utilization		
	CULTURAL CHARACTERISTICS			
	Colonies: <i>Nutrient Agar:</i> <i>Blood Agar:</i> Agar Slant: Nutrient Broth: Gelatin Stab: Oxygen Requirements: Optimum Temp.:	Nitrate Reduction		
H ₂ S Production				
Urease				
Catalase				
Oxidase				
DNase				
Phenylalanase				
Litmus Milk		REACTION		TIME
		Acid		_____
		Alkaline		_____
		Coagulation		_____
		Reduction		_____
		Peptonization		_____
No Change		_____		

Descriptive Chart			
STUDENT: _____		Habitat: _____ Culture No.: _____	
LAB SECTION: _____		Source: _____	
		Organism: _____	
MORPHOLOGICAL CHARACTERISTICS		PHYSIOLOGICAL CHARACTERISTICS	
Cell Shape: Arrangement: Size: Spores: Gram's Stain: Motility: Capsules: Special Stains:	Fermentation	TESTS	RESULTS
		Glucose	
		Lactose	
		Sucrose	
	Mannitol		
	Hydrolysis	Gelatin Liquefaction	
		Starch	
		Casein	
		Fat	
	IMViC	Indole	
		Methyl Red	
		V-P (acetylmethylcarbinol)	
Citrate Utilization			
CULTURAL CHARACTERISTICS			
Colonies: <i>Nutrient Agar:</i> <i>Blood Agar:</i> Agar Slant: Nutrient Broth: Gelatin Stab: Oxygen Requirements: Optimum Temp.:		Nitrate Reduction	
		H ₂ S Production	
		Urease	
		Catalase	
		Oxidase	
		DNase	
		Phenylalanase	
		Litmus Milk	REACTION
	Acid		
	Alkaline		
	Coagulation		
	Reduction		
	Peptonization		
No Change			

Descriptive Chart			
STUDENT: _____		Habitat: _____ Culture No.: _____	
LAB SECTION: _____		Source: _____	
		Organism: _____	
MORPHOLOGICAL CHARACTERISTICS		PHYSIOLOGICAL CHARACTERISTICS	
Cell Shape: Arrangement: Size: Spores: Gram's Stain: Motility: Capsules: Special Stains:		TESTS	RESULTS
	Fermentation	Glucose	
		Lactose	
		Sucrose	
		Mannitol	
	Hydrolysis	Gelatin Liquefaction	
		Starch	
		Casein	
		Fat	
	IMViC	Indole	
		Methyl Red	
		V-P (acetylmethylcarbinol)	
		Citrate Utilization	
		Nitrate Reduction	
	CULTURAL CHARACTERISTICS		
Colonies: <i>Nutrient Agar:</i> <i>Blood Agar:</i> Agar Slant: Nutrient Broth: Gelatin Stab: Oxygen Requirements: Optimum Temp.:		H ₂ S Production	
		Urease	
		Catalase	
		Oxidase	
		DNase	
		Phenylalanase	
	Litmus Milk	REACTION	TIME
		Acid	_____
		Alkaline	_____
		Coagulation	_____
		Reduction	_____
		Peptonization	_____
		No Change	_____

Descriptive Chart

STUDENT: _____		Habitat: _____ Culture No.: _____		
LAB SECTION: _____		Source: _____		
		Organism: _____		
MORPHOLOGICAL CHARACTERISTICS		PHYSIOLOGICAL CHARACTERISTICS		
Cell Shape: Arrangement: Size: Spores: Gram's Stain: Motility: Capsules: Special Stains:	TESTS		RESULTS	
	Fermentation	Glucose		_____
		Lactose		_____
		Sucrose		_____
		Mannitol		_____
	Hydrolysis	Gelatin Liquefaction		_____
		Starch		_____
		Casein		_____
		Fat		_____
	IMVIC	Indole		_____
		Methyl Red		_____
		V-P (acetylmethylcarbinol)		_____
		Citrate Utilization		_____
	CULTURAL CHARACTERISTICS			
	Colonies: <i>Nutrient Agar:</i> <i>Blood Agar:</i> Agar Slant: Nutrient Broth: Gelatin Stab: Oxygen Requirements: Optimum Temp.:	Nitrate Reduction		_____
H ₂ S Production		_____		
Urease		_____		
Catalase		_____		
Oxidase		_____		
DNase		_____		
Phenylalanase		_____		
Litmus Milk		REACTION		TIME
		Acid		_____
		Alkaline		_____
		Coagulation		_____
		Reduction		_____
		Peptonization		_____
No Change		_____		

Descriptive Chart

STUDENT: _____		Habitat: _____ Culture No.: _____		
LAB SECTION: _____		Source: _____		
		Organism: _____		
MORPHOLOGICAL CHARACTERISTICS		PHYSIOLOGICAL CHARACTERISTICS		
Cell Shape: Arrangement: Size: Spores: Gram's Stain: Motility: Capsules: Special Stains:	TESTS		RESULTS	
	Fermentation	Glucose	_____	
		Lactose	_____	
		Sucrose	_____	
		Mannitol	_____	
	Hydrolysis	Gelatin Liquefaction	_____	
		Starch	_____	
		Casein	_____	
		Fat	_____	
	IMViC	Indole	_____	
		Methyl Red	_____	
		V-P (acetylmethylcarbinol)	_____	
		Citrate Utilization	_____	
	CULTURAL CHARACTERISTICS			
	Colonies: <i>Nutrient Agar:</i> <i>Blood Agar:</i> Agar Slant: Nutrient Broth: Gelatin Stab: Oxygen Requirements: Optimum Temp.:	Nitrate Reduction		_____
H ₂ S Production		_____		
Urease		_____		
Catalase		_____		
Oxidase		_____		
DNase		_____		
Phenylalanase		_____		

Litmus Milk		REACTION	TIME	
		Acid	_____	
		Alkaline	_____	
		Coagulation	_____	
		Reduction	_____	
		Peptonization	_____	
No Change		_____		

A

Appendix
Tables

Table I International Atomic Weights

Element	Symbol	Atomic Number	Atomic Weight
Aluminum	Al	13	26.97
Antimony	Sb	51	121.76
Arsenic	As	33	74.91
Barium	Ba	56	137.36
Beryllium	Be	4	9.013
Bismuth	Bi	83	209.00
Boron	B	5	10.82
Bromine	Br	35	79.916
Cadmium	Cd	48	112.41
Calcium	Ca	20	40.08
Carbon	C	6	12.010
Chlorine	Cl	17	35.457
Chromium	Cr	24	52.01
Cobalt	Co	27	58.94
Copper	Cu	29	63.54
Fluorine	F	9	19.00
Gold	Au	79	197.2
Hydrogen	H	1	1.0080
Iodine	I	53	126.92
Iron	Fe	26	55.85
Lead	Pb	82	207.21
Magnesium	Mg	12	24.32
Manganese	Mn	25	54.93
Mercury	Hg	80	200.61
Nickel	Ni	28	58.69
Nitrogen	N	7	14.008
Oxygen	O	8	16.0000
Palladium	Pd	46	106.7
Phosphorus	P	15	30.98
Platinum	Pt	78	195.23
Potassium	K	19	39.096
Radium	Ra	88	226.05
Selenium	Se	34	78.96
Silicon	Si	14	28.06
Silver	Ag	47	107.880
Sodium	Na	11	22.997
Strontium	Sr	38	87.63
Sulfur	S	16	32.066
Tin	Sn	50	118.70
Titanium	Ti	22	47.90
Tungsten	W	74	183.92
Uranium	U	92	238.07
Vanadium	V	23	50.95
Zinc	Zn	30	65.38
Zirconium	Zr	40	91.22

Appendix A • Tables

Table II Four-Place Logarithms

N	0	1	2	3	4	5	6	7	8	9
10	0000	0043	0086	0128	0170	0212	0253	0294	0334	0374
11	0414	0453	0492	0531	0569	0607	0645	0682	0719	0755
12	0792	0828	0864	0899	0934	0969	1004	1038	1072	1106
13	1139	1173	1206	1239	1271	1303	1335	1367	1399	1430
14	1461	1492	1523	1553	1584	1614	1644	1673	1703	1732
15	1761	1790	1818	1847	1875	1903	1931	1959	1987	2014
16	2041	2068	2095	2122	2148	2175	2201	2227	2253	2279
17	2304	2330	2355	2380	2405	2430	2455	2480	2504	2529
18	2553	2577	2601	2625	2648	2672	2695	2718	2742	2765
19	2788	2810	2833	2856	2878	2900	2923	2945	2967	2989
20	3010	3032	3054	3075	3096	3118	3139	3160	3181	3201
21	3222	3243	3263	3284	3304	3324	3345	3365	3385	3404
22	3424	3444	3464	3483	3502	3522	3541	3560	3579	3598
23	3617	3636	3655	3674	3692	3711	3729	3747	3766	3784
24	3802	3820	3838	3856	3874	3892	3909	3927	3945	3962
25	3979	3997	4014	4031	4048	4065	4082	4099	4116	4133
26	4150	4166	4183	4200	4216	4232	4249	4265	4281	4298
27	4314	4330	4346	4362	4378	4393	4409	4425	4440	4456
28	4472	4487	4502	4518	4533	4548	4564	4579	4594	4609
29	4624	4639	4654	4669	4683	4698	4713	4728	4742	4757
30	4771	4786	4800	4814	4829	4843	4857	4871	4886	4900
31	4914	4928	4942	4955	4969	4983	4997	5011	5024	5038
32	5051	5065	5079	5092	5105	5119	5132	5145	5159	5172
33	5185	5198	5211	5224	5237	5250	5263	5276	5289	5302
34	5315	5328	5340	5353	5366	5378	5391	5403	5416	5428
35	5441	5453	5465	5478	5490	5502	5514	5527	5539	5551
36	5563	5575	5587	5599	5611	5623	5635	5647	5658	5670
37	5682	5694	5705	5717	5729	5740	5752	5763	5775	5786
38	5798	5809	5821	5832	5843	5855	5866	5877	5888	5899
39	5911	5922	5933	5944	5955	5966	5977	5988	5999	6010
40	6021	6031	6042	6053	6064	6075	6085	6096	6107	6117
41	6128	6138	6149	6160	6170	6180	6191	6201	6212	6222
42	6232	6243	6253	6263	6274	6284	6294	6304	6314	6325
43	6335	6345	6355	6365	6375	6385	6395	6405	6415	6425
44	6435	6444	6454	6464	6474	6484	6493	6503	6513	6522
45	6532	6542	6551	6561	6571	6580	6590	6599	6609	6618
46	6628	6637	6646	6656	6665	6675	6684	6693	6702	6712
47	6721	6730	6739	6749	6758	6767	6776	6785	6794	6803
48	6812	6821	6830	6839	6848	6857	6866	6875	6884	6893
49	6902	6911	6920	6928	6937	6946	6955	6964	6972	6981
50	6990	6998	7007	7016	7024	7033	7042	7050	7059	7067
51	7076	7084	7093	7101	7110	7118	7126	7135	7143	7152
52	7160	7168	7177	7185	7193	7202	7210	7218	7226	7235
53	7243	7251	7259	7267	7275	7284	7292	7300	7308	7316
54	7324	7332	7340	7348	7356	7364	7372	7380	7388	7396
N	0	1	2	3	4	5	6	7	8	9

Tables • **Appendix A**

Table II (continued)

N	0	1	2	3	4	5	6	7	8	9
55	7404	7412	7419	7427	7435	7443	7451	7459	7466	7474
56	7482	7490	7497	7505	7513	7520	7528	7536	7543	7551
57	7559	7566	7574	7582	7589	7597	7604	7612	7619	7627
58	7634	7642	7649	7657	7664	7672	7679	7686	7694	7701
59	7709	7716	7723	7731	7738	7745	7752	7760	7767	7774
60	7782	7789	7796	7803	7810	7818	7825	7832	7839	7846
61	7853	7860	7868	7875	7882	7889	7896	7903	7910	7917
62	7924	7931	7938	7945	7952	7959	7966	7973	7980	7987
63	7993	8000	8007	8014	8021	8028	8035	8041	8048	8055
64	8062	8069	8075	8082	8089	8096	8102	8109	8116	8122
65	8129	8136	8142	8149	8156	8162	8169	8176	8182	8189
66	8195	8202	8209	8215	8222	8228	8235	8241	8248	8254
67	8261	8267	8274	8280	8287	8293	8299	8306	8312	8319
68	8325	8331	8338	8344	8351	8357	8363	8370	8376	8382
69	8388	8395	8401	8407	8414	8420	8426	8432	8439	8445
70	8451	8457	8463	8470	8476	8482	8488	8494	8500	8506
71	8513	8519	8525	8531	8537	8543	8549	8555	8561	8567
72	8573	8579	8585	8591	8597	8603	8609	8615	8621	8627
73	8633	8639	8645	8651	8657	8663	8669	8675	8681	8686
74	8692	8698	8704	8710	8716	8722	8727	8733	8739	8745
75	8751	8756	8762	8768	8774	8779	8785	8791	8797	8802
76	8808	8814	8820	8825	8831	8837	8842	8848	8854	8859
77	8865	8871	8876	8882	8887	8893	8899	8904	8910	8915
78	8921	8927	8932	8938	8943	8949	8954	8960	8965	8971
79	8976	8982	8987	8993	8998	9004	9009	9015	9020	9025
80	9031	9036	9042	9047	9053	9058	9063	9069	9074	9079
81	9085	9090	9096	9101	9106	9112	9117	9122	9128	9133
82	9138	9143	9149	9154	9159	9165	9170	9175	9180	9186
83	9191	9196	9201	9206	9212	9217	9222	9227	9232	9238
84	9243	9248	9253	9258	9263	9269	9274	9279	9284	9289
85	9294	9299	9304	9309	9315	9320	9325	9330	9335	9340
86	9345	9350	9355	9360	9365	9370	9375	9380	9385	9390
87	9395	9400	9405	9410	9415	9420	9425	9430	9435	9440
88	9445	9450	9455	9460	9465	9469	9474	9479	9484	9489
89	9494	9499	9504	9509	9513	9518	9523	9528	9533	9538
90	9542	9547	9552	9557	9562	9566	9571	9576	9581	9586
91	9590	9595	9600	9605	9609	9614	9619	9624	9628	9633
92	9638	9643	9647	9652	9657	9661	9666	9671	9675	9680
93	9685	9689	9694	9699	9703	9708	9713	9717	9722	9727
94	9731	9736	9741	9745	9750	9754	9759	9763	9768	9773
95	9777	9782	9786	9791	9795	9800	9805	9809	9814	9818
96	9823	9827	9832	9836	9841	9845	9850	9854	9859	9863
97	9868	9872	9877	9881	9886	9890	9894	9899	9903	9908
98	9912	9917	9921	9926	9930	9934	9939	9943	9948	9952
99	9956	9961	9965	9969	9974	9978	9983	9987	9991	9996
100	0000	0004	0009	0013	0017	0022	0026	0030	0035	0039
N	0	1	2	3	4	5	6	7	8	9

Appendix A • Tables

Table III Temperature Conversion Table Centigrade to Fahrenheit

°C	0	1	2	3	4	5	6	7	8	9
−50	−58.0	−59.8	−61.6	−63.4	−65.2	−67.0	−68.8	−70.6	−72.4	−74.2
−40	−40.0	−41.8	−43.6	−45.4	−47.2	−49.0	−50.8	−52.6	−54.4	−56.2
−30	−22.0	−23.8	−25.6	−27.4	−29.2	−31.0	−32.8	−34.6	−36.4	−38.2
−20	− 4.0	− 5.8	− 7.6	− 9.4	−11.2	−13.0	−14.8	−16.6	−18.4	−20.2
−10	+14.0	+12.2	+10.4	+ 8.6	+ 6.8	+ 5.0	+ 3.2	+ 1.4	− 0.4	− 2.2
− 0	+32.0	+30.2	+28.4	+26.6	+24.8	+23.0	+21.2	+19.4	+17.6	+15.8
0	32.0	33.8	35.6	37.4	39.2	41.0	42.8	44.6	46.4	48.2
10	50.0	51.8	53.6	55.4	57.2	59.0	60.8	62.6	64.4	66.2
20	68.0	69.8	71.6	73.4	75.2	77.0	78.8	80.6	82.4	84.2
30	86.0	87.8	89.6	91.4	93.2	95.0	96.8	98.6	100.4	102.2
40	104.0	105.8	107.6	109.4	111.2	113.0	114.8	116.6	118.4	120.2
50	122.0	123.8	125.6	127.4	129.2	131.0	132.8	134.6	136.4	138.2
60	140.0	141.8	143.6	145.4	147.2	149.0	150.8	152.6	154.4	156.2
70	158.0	159.8	161.6	163.4	165.2	167.0	168.8	170.6	172.4	174.2
80	176.0	177.8	179.6	181.4	183.2	185.0	186.8	188.6	190.4	192.2
90	194.0	195.8	197.6	199.4	201.2	203.0	204.8	206.6	208.4	210.2
100	212.0	213.8	215.6	217.4	219.2	221.0	222.8	224.6	226.4	228.2
110	230.0	231.8	233.6	235.4	237.2	239.0	240.8	242.6	244.4	246.2
120	248.0	249.8	251.6	253.4	255.2	257.0	258.8	260.6	262.4	264.2
130	266.0	267.8	269.6	271.4	273.2	275.0	276.8	278.6	280.4	282.2
140	284.0	285.8	287.6	289.4	291.2	293.0	294.8	296.6	298.4	300.2
150	302.0	303.8	305.6	307.4	309.2	311.0	312.8	314.6	316.4	318.2
160	320.0	321.8	323.6	325.4	327.2	329.0	330.8	332.6	334.4	336.2
170	338.0	339.8	341.6	343.4	345.2	347.0	348.8	350.6	352.4	354.2
180	356.0	357.8	359.6	361.4	363.2	365.0	366.8	368.6	370.4	372.2
190	374.0	375.8	377.6	379.4	381.2	383.0	384.8	386.6	388.4	390.2
200	392.0	393.8	395.6	397.4	399.2	401.0	402.8	404.6	406.4	408.2
210	410.0	411.8	413.6	415.4	417.2	419.0	420.8	422.6	424.4	426.2
220	428.0	429.8	431.6	433.4	435.2	437.0	438.8	440.6	442.4	444.2
230	446.0	447.8	449.6	451.4	453.2	455.0	456.8	458.6	460.4	462.2
240	464.0	465.8	467.6	469.4	471.2	473.0	474.8	476.6	478.4	480.2
250	482.0	483.8	485.6	487.4	489.2	491.0	492.8	494.6	496.4	498.2

$$^{\circ}\text{F} = ^{\circ}\text{C} \times 9/5 + 32$$

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$$

Tables • **Appendix A**

Table IV Autoclave Steam Pressures and Corresponding Temperatures

Steam Pressure lb/sq in	Temperature		Steam Pressure lb/sq in	Temperature		Steam Pressure lb/sq in	Temperature	
	°C	°F		°C	°F		°C	°F
0	100.0	212.0						
1	101.9	215.4	11	116.4	241.5	21	126.9	260.4
2	103.6	218.5	12	117.6	243.7	22	127.8	262.0
3	105.3	221.5	13	118.8	245.8	23	128.7	263.7
4	106.9	224.4	14	119.9	247.8	24	129.6	265.3
5	108.4	227.1	15	121.0	249.8	25	130.4	266.7
6	109.8	229.6	16	122.0	251.6	26	131.3	268.3
7	111.3	232.3	17	123.0	253.4	27	132.1	269.8
8	112.6	234.7	18	124.1	255.4	28	132.9	271.2
9	113.9	237.0	19	125.0	257.0	29	133.7	272.7
10	115.2	239.4	20	126.0	258.8	30	134.5	274.1

Figures are for steam pressure only and the presence of any air in the autoclave invalidates temperature readings from the above table.

Appendix A • Tables

Table V Autoclave Temperatures as Related to the Presence of Air

Gauge Pressure, lb	Pure steam, complete air discharge		Two-thirds air discharge, 20-in. vacuum		One-half air discharge, 15-in. vacuum		One-third air discharge, 10-in. vacuum		No air discharge	
	°C	°F	°C	°F	°C	°F	°C	°F	°C	°F
5	109	228	100	212	94	202	90	193	72	162
10	115	240	109	228	105	220	100	212	90	193
15	121	250	115	240	112	234	109	228	100	212
20	126	259	121	250	118	245	115	240	109	228
25	130	267	126	259	124	254	121	250	115	240
30	135	275	130	267	128	263	126	259	121	250

Tables • **Appendix A**

Table VI MPN Determination from Multiple Tube Test

NUMBER OF TUBES GIVING POSITIVE REACTION OUT OF			MPN Index per 100 ml	95 PERCENT CONFIDENCE LIMITS	
3 of 10 ml each	3 of 1 ml each	3 of 0.1 ml each		Lower	Upper
0	0	1	3	<0.5	9
0	1	0	3	<0.5	13
1	0	0	4	<0.5	20
1	0	1	7	1	21
1	1	0	7	1	23
1	1	1	11	3	36
1	2	0	11	3	36
2	0	0	9	1	36
2	0	1	14	3	37
2	1	0	15	3	44
2	1	1	20	7	89
2	2	0	21	4	47
2	2	1	28	10	150
3	0	0	23	4	120
3	0	1	39	7	130
3	0	2	64	15	380
3	1	0	43	7	210
3	1	1	75	14	230
3	1	2	120	30	380
3	2	0	93	15	380
3	2	1	150	30	440
3	2	2	210	35	470
3	3	0	240	36	1,300
3	3	1	460	71	2,400
3	3	2	1,100	150	4,800

From *Standard Methods for the Examination of Water and Wastewater*, Twelfth edition. (New York: The American Public Health Association, Inc.), p. 608.

Appendix A • Tables

Table VII Significance of zones of inhibition in Kirby-Bauer Method of antimicrobial sensitivity testing (1995)

Antimicrobial Agent	Disk Potency	R Resistant mm	I Intermediate mm	S Sensitive mm
Amikacin	30 mcg	<14	15–16	>17
Amoxicillin/Clavulanic Acid	30 mcg			
Staphylococci		<19	14–17	>20
Other gram-positive organisms		<13	14–17	>18
Ampicillin	75 mcg			
Gram-negative enterics		<13	14–16	>17
Staphylococci		<28		>29
Enterococci		<16		>17
Streptococci (not <i>S. pneumoniae</i>)		<21	22–29	>30
<i>Haemophilus</i> spp.		<18	19–21	>22
<i>Listeria monocytogenes</i>		<19		>20
Azlocillin (<i>Pseudomonas aeruginosa</i>)	75 mcg	<17		>18
Carbenicillin (<i>P. aeruginosa</i>)	100 mcg	<13	14–16	>17
Other gram-negative organisms		<19	20–22	>23
Cefactor	30 mcg	<14	15–17	>18
Cephalothin	30 mcg	<14	15–17	>18
Chloramphenicol	30 mcg	<12	13–17	>18
<i>S. pneumoniae</i>		<20		>21
Clarithromycin	15 mcg	<13	14–17	>18
<i>S. pneumoniae</i>		<16	17–20	>21
Clindamycin	2 mcg	<14	15–20	>21
<i>S. pneumoniae</i>		<20		>21
Erythromycin	15 mcg	<13	14–22	>23
<i>S. pneumoniae</i>		<15	16–20	>21
Gentamicin	10 mcg	<12	13–14	>15
Imipenem	10 mcg	<13	14–15	>16
<i>Haemophilus</i> spp.				>16
Kanamycin	30 mcg	<13	14–17	>18
Lomefloxacin	10 mcg	<18	19–21	>22
Loracarbef	30 mcg	<14	15–17	>18
Mezlocillin (<i>P. aeruginosa</i>)	75 mcg	<15		>16
Other gram-negative organisms		<17	18–20	>19
Mincycline	30 mcg	<14	15–18	>19
Moxalactam	30 mcg	<14	15–22	>23
Nafcillin	1 mcg	<10	11–12	>13
Nalidixic Acid	30 mcg	<13	14–18	>19
Netilmicin	30 mcg	<12	13–14	>15
Norfloxacin	10 mcg	<12	13–16	>17
Ofloxacin	5 mcg	<12	13–15	>16
Penicillin G (Staphylococci)	10 units	<28		>29
Enterococci		<14		>15
Streptococci (not <i>S. pneumoniae</i>)		<19	20–27	>28
<i>Neisseria gonorrhoeae</i>		<26	27–46	>47
<i>L. monocytogenes</i>		<19		>20
Piperacillin/Tazobactam	100/10 mcg			
Staphylococci		<17		>18
<i>P. aeruginosa</i>		<17		>18
Other gram-negative organisms		<14	15–19	>20
Rifampin	5 mcg	<16	17–19	>20
<i>Haemophilus</i> spp.		<16	17–19	>20
<i>S. pneumoniae</i>		<16	17–18	>19
Streptomycin	10 mcg	<11	12–14	>15
Sulfisoxazole	300 mcg	<12	13–16	>17
Tetracycline	30 mcg	<14	15–18	>19
<i>S. pneumoniae</i>		<17	18–21	>22
Tobramycin	10 mcg	<12	13–14	>15
Trimethoprim/Sulfamethoxazole	1.25/23.75	<10	11–15	>16
Vancomycin	30 mcg	<14	15–16	>17

Table VIII Indicators of Hydrogen Ion Concentration

Many of the following indicators are used in the media of certain exercises in this manual. This table indicates the pH range of each indicator and the color changes that occur. To determine the exact pH within a particular range one should use a set of standard colorimetric tubes that are available from the prep room. Consult your lab instructor.

Indicator	Full Acid Color	Full Alkaline Color	pH Range
Cresol Red	red	yellow	0.2 – 1.8
Metacresol Purple (acid range)	red	yellow	1.2 – 2.8
Thymol Blue	red	yellow	1.2 – 2.8
Bromphenol Blue	yellow	blue	3.0 – 4.6
Bromcresol Green	yellow	blue	3.8 – 5.4
Chlorcresol Green	yellow	blue	4.0 – 5.6
Methyl Red	red	yellow	4.4 – 6.4
Chlorphenol Red	yellow	red	4.8 – 6.4
Bromcresol Purple	yellow	purple	5.2 – 6.8
Bromthymol Blue	yellow	blue	6.0 – 7.6
Neutral Red	red	amber	6.8 – 8.0
Phenol Red	yellow	red	6.8 – 8.4
Cresol Red	yellow	red	7.2 – 8.8
Metacresol Purple (alkaline range)	yellow	purple	7.4 – 9.0
Thymol Blue (alkaline range)	yellow	blue	8.0 – 9.6
Cresolphthalein	colorless	red	8.2 – 9.8
Phenolphthalein	colorless	red	8.3 – 10.0

B

Appendix
Indicators, Stains, Reagents

INDICATORS

All the indicators used in this manual can be made by (1) dissolving a measured amount of the indicator in 95% ethanol, (2) adding a measured amount of water, and (3) filtering with filter paper. The following chart provides the correct amounts of indicator, alcohol, and water for various indicator solutions.

Indicator Solution	Indicator (gm)	95% Ethanol (ml)	Distilled H ₂ O (ml)
Bromcresol green	0.4	500	500
Bromcresol purple	0.4	500	500
Bromthymol blue	0.4	500	500
Cresol red	0.2	500	500
Methyl red	0.2	500	500
Phenolphthalein	1.0	50	50
Phenol red	0.2	500	500
Thymol blue	0.4	500	500

STAINS AND REAGENTS

Acid-Alcohol (for Ziehl-Neelsen stain)

3 ml concentrated hydrochloric acid in 100 ml of 95% ethyl alcohol.

Acid-Alcohol (for fluorochrome staining)

HCl2.5 ml
Ethyl alcohol, 70%500.0 ml
NaCl2.5 gm

Alcohol, 70% (from 95%)

Alcohol, 95%368.0 ml
Distilled water132.0 ml

Auramine-Rhodamine Stain

(for mycobacteria)

Auramine O3.0 gm
Rhodamine B1.5 gm
Glycerol150.0 ml
Phenol crystals (liquefied at 50° C) ...20.0 ml
Distilled water100.0 ml

Clarify by filtration with glass wool or Whatman #2 filter paper. Do not heat. Store at room temperature.

Barritt's Reagent (Voges-Proskauer test)

Solution A: 6 gm alpha-naphthol in 100 ml 95% ethyl alcohol.

Solution B: 16 gm potassium hydroxide in 100 ml water.

Note that no creatine is used in these reagents as is used in O'Meara's reagent for the V-P test.

Carbolfuchsin Stain (Ziehl's)

Solution A: Dissolve 0.3 gm of basic fuchsin (90% dye content) in 10 ml 95% ethyl alcohol.

Solution B: Dissolve 5 gm of phenol in 95 ml of water.

Mix solutions A and B.

Crystal Violet Stain (Hucker modification)

Solution A: Dissolve 2.0 gm of crystal violet (85% dye content) in 20 ml of 95% ethyl alcohol.

Solution B: Dissolve 0.8 gm ammonium oxalate in 80.0 ml distilled water.

Mix solutions A and B.

Appendix B • Indicators, Stains, Reagents

Diphenylamine Reagent (nitrate test)

Dissolve 0.7 gm diphenylamine in a mixture of 60 ml of concentrated sulfuric acid and 28.8 ml of distilled water.

Cool and add slowly 11.3 ml of concentrated hydrochloric acid. After the solution has stood for 12 hours some of the base separates, showing that the reagent is saturated.

Ferric Chloride Reagent (Ex. 77)

FeCl₃·6H₂O12 gm
2% Aqueous HCl100 ml

Make up the 2% aq. HCl by adding 5.4 ml of concentrated HCl (37%) to 94.6 ml H₂O. Inoculate with two or three colonies of beta hemolytic streptococci, incubate at 35° C for 20 or more hours. Centrifuge the medium to pack the cells, and pipette 0.8 ml of the clear supernate into a Kahn tube. Add 0.2 ml of the ferric chloride reagent to the Kahn tube and mix well. If a heavy precipitate remains longer than 10 minutes, the test is positive.

Gram's Iodine (Lugol's)

Dissolve 2.0 gm of potassium iodide in 300 ml of distilled water and then add 1.0 gm iodine crystals.

Iodine, 5% Aqueous Solution (Ex. 41)

Dissolve 4 gm of potassium iodide in 300 ml of distilled water and then add 2.0 gm iodine crystals.

Kovacs' Reagent (indole test)

n-amyl alcohol75.0 ml
Hydrochloric acid (conc.)25.0 ml
p-dimethylamine-benzaldehyde5.0 gm

Lactophenol Cotton Blue Stain

Phenol crystals20 gm
Lactic acid20 ml
Glycerol40 ml
Cotton blue0.05 gm

Dissolve the phenol crystals in the other ingredients by heating the mixture gently under a hot water tap.

Leifson's Flagellar Stain

Prepare three separate solutions as follows:

Solution A: Dissolve 0.9 gm of pararosaniline acetate and 0.3 gm of pararosaniline hydrochloride

in 100 ml of 95% ethyl alcohol. Let stand overnight at room temperature to complete dissolution.

Solution B: Dissolve 3 gm of tannic acid in 100 ml of distilled water.

Solution C: Dissolve 1.5 gm of sodium chloride in 100 ml of distilled water.

Mix equal volumes of solutions A, B, and C and let stand for 2 hours. Store in a stoppered bottle in a refrigerator (up to 2 months). Disregard precipitate that forms in bottom of bottle. Do not filter. Will store indefinitely, if frozen. Frozen stain solution must be thoroughly mixed after thawing since the water separates from the alcohol. After mixing, the precipitate should be allowed to settle to the bottom.

Note: Pararosaniline compounds should be certified for flagellar staining.

Malachite Green Solution (spore stain)

Dissolve 5.0 gm malachite green oxalate in 100 ml distilled water.

McFarland Nephelometer Barium Sulfate Standards (Ex. 55)

Prepare 1% aqueous barium chloride and 1% aqueous sulfuric acid solutions.

Add the amounts indicated in table 1 to clean, dry ampoules. Ampoules should have the same diameter as the test tube to be used in subsequent density determinations.

Seal the ampoules and label them.

Table 1 Amounts for Standards

Tube	Barium Chloride 1% (ml)	Sulfuric Acid 1% (ml)	Corresponding Approx. Density of Bacteria (million/ml)
1	0.1	9.9	300
2	0.2	9.8	600
3	0.3	9.7	900
4	0.4	9.6	1200
5	0.5	9.5	1500
6	0.6	9.4	1800
7	0.7	9.3	2100
8	0.8	9.2	2400
9	0.9	9.1	2700
10	1.0	9.0	3000

Indicators, Stains, Reagents • **Appendix B**

Methylene Blue (Loeffler's)

Solution A: Dissolve 0.3 gm of methylene blue (90% dye content) in 30.0 ml ethyl alcohol (95%).

Solution B: Dissolve 0.01 gm potassium hydroxide in 100.0 ml distilled water. Mix solutions A and B.

Naphthol, alpha

5% alpha-naphthol in 95% ethyl alcohol

Caution: Avoid all contact with human tissues. Alpha-naphthol is considered to be carcinogenic.

Nessler's Reagent (ammonia test)

Dissolve about 50 gm of potassium iodide in 35 ml of cold ammonia-free distilled water. Add a saturated solution of mercuric chloride until a slight precipitate persists. Add 400 ml of a 50% solution of potassium hydroxide. Dilute to 1 liter, allow to settle, and decant the supernatant for use.

Nigrosine Solution (Dorner's)

Nigrosine, water soluble10 gm
Distilled water100 ml

Boil for 30 minutes. Add as a preservative 0.5 ml formaldehyde (40%). Filter twice through double filter paper and store under aseptic conditions.

Nitrate Test Reagent (see Diphenylamine)

Nitrite Test Reagents

Solution A: Dissolve 8 gm sulfanilic acid in 1000 ml 5N acetic acid (1 part glacial acetic acid to 2.5 parts water).

Solution B: Dissolve 5 gm dimethyl-alpha-naphthylamine in 1000 ml 5N acetic acid. Do not mix solutions.

Caution: Although at this time it is not known for sure, there is a possibility that dimethyl- α -naphthylamine in solution B may be carcinogenic. For reasons of safety, avoid all contact with tissues.

Oxidase Test Reagent

Mix 1.0 gm of dimethyl-p-phenylenediamine hydrochloride in 100 ml of distilled water.

Preferably, the reagent should be made up fresh, daily. It should not be stored longer than one week in the refrigerator. Tetramethyl-p-phenylenediamine dihydrochloride (1%) is even more sensitive, but is considerably more expensive and more difficult to obtain.

Phenolized Saline

Dissolve 8.5 gm sodium chloride and 5.0 gm phenol in 1 liter distilled water.

Physiological Saline

Dissolve 8.5 gm sodium chloride in 1 liter distilled water.

Potassium permanganate (for fluorochrome staining)

KMnO₄2.5 gm
Distilled water500.0 ml

Safranin (for gram staining)

Safranin O (2.5% sol'n in 95% ethyl alcohol)10.0 ml
Distilled water100.0 ml

Trommsdorf's Reagent (nitrite test)

Add slowly, with constant stirring, 100 ml of a 20% aqueous zinc chloride solution to a mixture of 4.0 gm of starch in water. Continue heating until the starch is dissolved as much as possible, and the solution is nearly clear. Dilute with water and add 2 gm of potassium iodide. Dilute to 1 liter, filter, and store in amber bottle.

Vaspar

Melt together 1 pound of Vaseline and 1 pound of paraffin. Store in small bottles for student use.

Voges-Proskauer Test Reagent (see Barritt's)

White Blood Cell (WBC) Diluting Fluid

Hydrochloric acid5 ml
Distilled water495 ml

Add 2 small crystals of thymol as a preservative.

C

Appendix Media

Conventional Media The following media are used in the experiments of this manual. All of these media are available in dehydrated form from either Difco Laboratories, Detroit, Michigan, or Baltimore Biological Laboratory (BBL), a division of Becton, Dickinson & Co., Cockeysville, Maryland. Compositions, methods of preparation, and usage will be found in their manuals, which are supplied upon request at no cost. The source of each medium is designated as (B) for BBL and (D) for Difco.

Bile esculin (D)	Nutrient broth (B,D)
Brewer's anaerobic agar (D)	Nutrient gelatin (B,D)
Desoxycholate citrate agar (B,D)	Phenol red sucrose broth (B,D)
Desoxycholate lactose agar (B,D)	Phenylalanine agar (D)
DNase test agar (B,D)	Phenylethyl alcohol medium (B)
Endo agar (B,D)	Russell double sugar agar (B,D)
Eugonagar (B,D)	Sabouraud's glucose (dextrose) agar (D)
Fluid thioglycollate medium (B,D)	Semisolid medium (B)
Heart infusion agar (D)	Simmons citrate agar (B,D)
Hektoen Enteric Agar (B,D)	Snyder test agar (D)
Kligler iron agar (B,D)	Sodium hippurate (D)
Lead acetate agar (D)	Spirit blue agar (D)
Levine EMB agar (B,D)	SS agar (B,D)
Lipase reagent (D)	<i>m</i> -Staphylococcus broth (D)
Litmus milk (B,D)	Staphylococcus medium 110 (D)
Lowenstein-Jensen medium (B,D)	Starch agar (D)
MacConkey Agar (B,D)	Trypticase soy agar (B)
Mannitol salt agar (B,D)	Trypticase soy broth (B)
MR-VP medium (D)	Tryptone glucose extract agar (B,D)
Mueller-Hinton medium (B,D)	Urea (urease test) broth (B,D)
Nitrate broth (D)	Veal infusion agar (B,D)
Nutrient agar (B,D)	Xylose Lysine Desoxycholate Agar (B,D)

Special Media The following media are not included in the manuals that are supplied by Difco and BBL; therefore, methods of preparation are presented here.

Ammonium Medium (for *Nitrosomonas*)

(NH ₄) ₂ SO ₄	2.0 gm
MgSO ₄ · 7H ₂ O	0.5 gm
FeSO ₄ · 7H ₂ O	0.03 gm
NaCl	0.3 gm
MgCO ₃	10.0 gm
K ₂ HPO ₄	1.0 gm
Water	1000.0 ml

For the isolation of *Nitrosomonas* from soil, sterilization is not necessary if the inoculations are made as soon as the medium is made up. The pH should be adjusted to 7.3. If sterilization is desirable for storage or other reasons, adjust the pH aseptically after sterilization with sterile 1N HCl.

Bile Esculin Slants (Ex. 79)

Heart infusion agar	40.0 gm
Esculin	1.0 gm
Ferric chloride	0.5 gm
Distilled water	1000.0 ml
Dispense into sterile 15 × 125 mm screw-capped tubes, sterilize in autoclave at 121° C for 15 minutes, and slant during cooling.	

Blood Agar

Trypticase soy agar powder	40 gm
Distilled water	1000 ml
Final pH of 7.3	
Defibrinated sheep or rabbit blood	50 ml

Appendix C • Media

Liquefy and sterilize 1000 ml of trypticase soy agar in a large Erlenmeyer flask. While the TSA is being sterilized, warm up 50 ml of defibrinated blood to 50° C. After cooling the TSA to 50° C, aseptically transfer the blood to the flask and mix by gently rotating the flask (cold blood may cause lumpiness).

Pour 10–12 ml of the mixture into sterile Petri plates. If bubbles form on the surface of the medium, flame the surface gently with a Bunsen burner before the medium solidifies. It is best to have an assistant to lift off the Petri plate lids while pouring the medium into the plates. A full flask of blood agar is somewhat cumbersome to handle with one hand.

Bromthymol Blue Carbohydrate Broths**Make up stock indicator solution:**

Bromthymol blue	.8 gm
95% ethyl alcohol	.250 ml
Distilled water	.250 ml

Indicator is dissolved first in alcohol and then water is added.

Make up broth:

Sugar base (lactose, sucrose, glucose, etc.)	.5 gm
Tryptone	.10 gm
Yeast extract	.5 gm
Indicator solution	.2 ml
Distilled water	1000 ml

Final pH 7.0

Deca-Strength Phage Broth (Ex. 28)

Peptone	100 gm
Yeast extract	50 gm
NaCl	25 gm
K ₂ HPO ₄	80 gm
Distilled water	1000 ml

Final pH 7.6

Emmons' Culture Medium for Fungi

C. W. Emmons developed the following recipe as an improvement over Sabouraud's glucose agar for the cultivation of fungi. Its principal advantage is that a neutral pH does not inhibit certain molds that have difficulty growing on Sabouraud's agar (pH 5.6). Instead of relying on

a low pH to inhibit bacteria, it contains chloramphenicol, which does not adversely affect the fungi.

Glucose	20 gm
Neopeptone	10 gm
Agar	20 gm
Chloramphenicol	40 mg
Distilled water	1000 ml

After the glucose, peptone, and agar are dissolved, heat to boiling, add the chloramphenicol which has been suspended in 10 ml of 95% alcohol and remove quickly from the heat. Autoclave for only 10 minutes.

Glucose–Minimal Salts Agar

(Ex. 76, Ames test)

This medium is made from glucose, agar, and Vogel-Bonner medium E (50×).

Vogel-Bonner Medium E (50×)

Distilled water (45° C)	.670 ml
MgSO ₄ · 7H ₂ O	.10 gm
Citric acid monohydrate	.100 gm
K ₂ HPO ₄ (anhydrous)	.500 gm
Sodium ammonium phosphate (NaH ₂ NH ₄ PO ₄ · 4H ₂ O)	.175 gm

Add salts in the order indicated to warm water (45° C) in a 2-liter beaker or flask placed on a magnetic stirring hot plate. Allow each salt to dissolve completely before adding the next. Adjust the volume to 1 liter. Distribute into two 1-liter glass bottles. Autoclave, loosely capped, for 20 minutes at 121° C.

Plates of Glucose–Minimal Salts Agar

Agar	15 gm
Distilled water	930 ml
50× V-B salts	20 ml
40% glucose	50 ml

Add 15 gm of agar to 930 ml of distilled water in a 2-liter flask. Autoclave for 20 minutes using slow exhaust. When the solution has cooled slightly, add 20 ml of sterile 50× V-B salts and 50 ml of sterile 40% glucose. For mixing, a large magnetic stir bar can be added to the flask before autoclaving. After all the ingredients have been added, the solution should be stirred thoroughly. Pour 30 ml into each Petri plate. **Important:** The 50× V-B salts and 40% glucose should be autoclaved separately.

Media • Appendix C

Glucose Peptone Acid Agar

Glucose	10 gm
Peptone	.5 gm
Monopotassium phosphate	.1 gm
Magnesium sulfate ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$)	.05 gm
Agar	.15 gm
Water	1000 ml

While still liquid after sterilization, add sufficient sulfuric acid to bring the pH down to 4.0.

Glycerol Yeast Extract Agar

Glycerol	.5 ml
Yeast extract	.2 gm
Dipotassium phosphate	.1 gm
Agar	.15 gm
Water	1000 ml

m Endo MF Broth (Ex. 64)

This medium is extremely hygroscopic in the dehydrated form and oxidizes quickly to cause deterioration of the medium after the bottle has been opened. Once a bottle has been opened it should be dated and discarded after one year. If the medium becomes hardened within that time it should be discarded. Storage of the bottle inside a larger bottle that contains silica gel will extend shelf life.

Failure of Exercise 64 can often be attributed to faulty preparation of the medium. It is best to make up the medium the day it is to be used. It should not be stored over 96 hours prior to use. The Millipore Corporation recommends the following method for preparing this medium. (These steps are not exactly as stated in the Millipore Application Manual AM302.)

1. Into a 250 ml screw-cap Erlenmeyer flask place the following:

Distilled water	.50 ml
95% ethyl alcohol	.2 ml
Dehydrated medium (<i>m</i> Endo MF broth)	.48 gm

Shake the above mixture by swirling the flask until the medium is dissolved and then add another 50 ml of distilled water.

2. Cap the flask loosely and immerse it into a pan of boiling water. As soon as the medium begins to simmer, remove the flask from the water bath. Do not boil the medium any further.

3. Cool the medium to 45° C, and adjust the pH to between 7.1 and 7.3.
4. If the medium must be stored for a few days, place it in the refrigerator at 2°–10° C, with screw-cap tightened securely.

Milk Salt Agar (15% NaCl)

Prepare three separate beakers of the following ingredients:

1. Beaker containing 200 grams of sodium chloride.
2. Large beaker (2000 ml size) containing 50 grams of skim milk powder in 500 ml of distilled water.
3. Glycerol-peptone agar medium:

$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$	.5.0 gm
$\text{MgNO}_3 \cdot 6\text{H}_2\text{O}$	.1.0 gm
$\text{FeCl}_3 \cdot 7\text{H}_2\text{O}$	.0.025 gm
Difco proteose-peptone #3	.5.0 gm
Glycerol	.10.0 gm
Agar	.30.0 gm
Distilled water	.500.0 ml

Sterilize the above three beakers separately. The milk solution should be sterilized at 113°–115° C (8 lb pressure) in autoclave for 20 minutes. The salt and glycerol-peptone agar can be sterilized at conventional pressure and temperature. After the milk solution has cooled to 55° C, add the sterile salt, which should also be cooled down to a moderate temperature. If the salt is too hot, coagulation may occur. Combine the milk-salt and glycerol-peptone agar solutions by gently swirling with a glass rod. Dispense aseptically into Petri plates.

Nitrate Broth

Beef extract	.3 gm
Peptone	.5 gm
Potassium nitrate	.1 gm
Distilled water	1000 ml

Final pH 7.0 at 25° C

Nitrate Agar

Beef extract	.3 gm
Peptone	.5 gm
Potassium nitrate	.1 gm
Agar	.12 gm
Distilled water	1000 ml

Final pH 6.8 at 25° C

Appendix C • Media

Nitrate Succinate–Mineral Salts Broth

(Ex. 62)

This medium is used as an enrichment medium for isolating denitrifying bacteria from soil. Note that two stock solutions (A and B) should be made up before attempting to put together the complete medium.

Solution A (Trace Mineral Salts)

FeSO ₄ · 7H ₂ O	300 mg
MnCl ₂ · 4H ₂ O	180 mg
Co(NO ₃) ₂ · 7H ₂ O	130 mg
ZnSO ₄ · 7H ₂ O	40 mg
H ₂ MoO ₄	20 mg
CuSO ₄ · 5H ₂ O	1 mg
CaCl ₂	1000 mg

This solution should be stored at 4° C until used.

Solution B

NH ₄ Cl	1 gm
Na ₂ HPO ₄	2.14 gm
KH ₂ PO ₄	1.09 gm
MgSO ₄ · 7H ₂ O	0.2 gm
Trace mineral salts (Sol A)	10 ml
Water to	1000 ml

Complete Medium

Solution B	1000 ml
Sodium succinate	2 gm
Potassium nitrate	3 gm

Adjust the pH to 6.8, dispense into bottles, and autoclave at standard conditions.

Nitrate Succinate–Mineral Salts Agar

(Ex. 62)

Add 15 g agar to 1000 ml of the above complete medium. Dispense into Petri plates and sterilize in the autoclave.

Nitrite Medium (for *Nitrobacter*)

NaNO ₂	1.0 gm
MgSO ₄ · 7H ₂ O	0.5 gm
FeSO ₄ · 7H ₂ O	0.03 gm
NaCl	0.3 gm
Na ₂ CO ₃	1.0 gm
K ₂ HPO ₄	1.0 gm
Water	1000.0 ml

For the isolation of *Nitrobacter* from soil, sterilization is not necessary if the inoculations are made as soon as the medium is made up. The pH should be adjusted to 7.3. If sterilization is desirable for storage or other reasons, adjust the pH aseptically after sterilization with sterile 1N HCl.

Nitrogen-Free Glucose Agar (Ex. 59)

Add 15 grams of agar to the basal salts portion of the above recipe, bring to a boil, and sterilize in the autoclave at 121° C for 15 minutes. The glucose is dissolved in 100 ml of water and sterilized separately in similar manner. Mix the two solutions aseptically, and dispense into sterile Petri plates.

Nitrogen-Free Medium(Ex. 59, *Azotobacter*)

K ₂ HPO ₄	1.0 gm
MgSO ₄ · 7H ₂ O	0.2 gm
FeSO ₄ · 7H ₂ O	0.05 gm
CaCl ₂ · 2H ₂ O	0.1 gm
Na ₂ MoO ₄ · 2H ₂ O	0.001 gm
*Glucose	10.0 gm
Distilled water	1000 ml
*Sterilize separately.	

Adjust pH to 7.2.

If this medium is to be used immediately to isolate *Azotobacter* from soil, sterilization is not necessary. When it must be stored for any length of time, it should be sterilized.

If it is to be sterilized, the glucose should be dissolved separately in 100 ml of water and sterilized at 121° C for 15 minutes. The remainder of the medium is sterilized in a similar manner.

After sterilization, the two solutions are mixed aseptically and dispensed into sterile 8-oz prescription bottles (50 ml per bottle).

Phage Growth Medium (Ex. 29)

KH ₂ PO ₄	1.5 gm
Na ₂ HPO ₄	3.0 gm
NH ₄ Cl	1.0 gm
MgSO ₄ · 7H ₂ O	0.2 gm
Glycerol	10.0 gm
Acid-hydrolyzed casein	5.0 gm
dl-Tryptophan	0.01 gm
Gelatin	0.02 gm
Tween-80	0.2 gm
Distilled water	1000.0 ml

Sterilize in autoclave at 121° C for 20 minutes.

Phage Lysing Medium (Ex. 29)

Add sufficient sodium cyanide (NaCN) to the above growth medium to bring the concentration up to 0.02M. For 1 liter of lysing medium this will amount to about 1 gram (actually 0.98 gm) of NaCN. When an equal amount of this lysing medium is added to the growth medium during the last 6 hours of incubation, the concentration of NaCN in the combined medium is 0.01M.

Rhodospirillaceae Medium (Ex. 27)

This culture medium is used for the enrichment and culture of anaerobic phototrophic bacteria. To make up this medium you need to first prepare three stock solutions (A, B, and C) before putting together the entire batch.

A. Iron Citrate Solution

Ammonium ferrous sulfate	748 mg
Sodium citrate	1180 mg
Water to	500 ml

Store this stock solution at 4° C until needed.

B. Vitamin B₁₂ Solution

Certain strains require this vitamin. To make up 100 ml of this solution add 1 mg to 100 ml of water. Store at 4° C until needed.

C. Trace Metals Solution

H ₃ BO ₄	2.86 gm
MnCl ₂ · H ₂ O	1.81 gm
ZnSO ₄ · 7H ₂ O	0.222 gm
Na ₂ MoO ₄ · 2H ₂ O	0.390 gm
CuSO ₄ · 5H ₂ O	0.079 gm
Co(NO ₃) ₂ · 6H ₂ O	0.0494 gm
Water to	1000 ml

Store at 4° C until needed.

Complete Enrichment Medium

The final batch of this medium has the following ingredients. The succinate provides the organic carbon and the yeast extract provides essential vitamins for certain strains.

KH ₂ PO ₄	0.5 gm
MgSO ₄ · 7H ₂ O	0.2 gm
NaCl	0.4 gm
NH ₄ Cl	0.4 gm
CaCl ₂ · 2H ₂ O	0.05 gm
Sodium succinate	1.0 gm
Yeast extract	0.2 gm
Iron citrate solution (A)	5 ml
Vitamin B ₁₂ solution (B)	0.1 ml
Trace elements solution (C)	1 ml
Water to	1000 ml

Adjust the pH to 6.8, dispense into bottles and autoclave at standard conditions.

Russell Double Sugar Agar (Ex. 80)

Beef extract	1 gm
Proteose Peptone No. 3 (Difco)	12 gm
Lactose	10 gm
Dextrose	1 gm
Sodium chloride	5 gm
Agar	15 gm
Phenol red (Difco)	0.025 gm
Distilled water	1000 ml

Final pH 7.5 at 25° C

Dissolve ingredients in water, and bring to boiling. Cool to 50°–60° C, and dispense about 8 ml per tube (16 mm dia tubes). Slant tubes to cool. Butt depth should be about ½".

Skim Milk Agar

Skim milk powder	100 gm
Agar	15 gm
Distilled water	1000 ml

Dissolve the 15 gm of agar into 700 ml of distilled water by boiling. Pour into a large flask and sterilize at 121° C, 15 lb pressure.

In a separate container, dissolve the 100 gm of skim milk powder into 300 ml of water heated to 50° C. Sterilize this milk solution at 113°–115° C (8 lb pressure) for 20 minutes.

After the two solutions have been sterilized, cool to 55° C and combine in one flask, swirling gently to avoid bubbles. Dispense into sterile Petri plates.

Appendix C • Media

Sodium Chloride (6.5%) Tolerance

Broth (Ex. 79)

Heart infusion broth	.25 gm
NaCl	.60 gm
Indicator (1.6 gm bromocresol purple in 100 ml 95% ethanol)	.1 ml
Dextrose	.1 gm
Distilled water	1000 ml

Add all reagents together up to 1000 ml (final volume). Dispense in 15×125 mm screw-capped tubes and sterilize in an autoclave 15 minutes at 121° C.

A positive reaction is recorded when the indicator changes from purple to yellow or when growth is obvious even though the indicator does not change.

Sodium Hippurate Broth (Ex. 79)

Heart infusion broth	.25 gm
Sodium hippurate	.10 gm
Distilled water	1000 ml

Sterilize in autoclave at 121° C for 15 minutes after dispensing in 15 × 125 mm screw-capped tubes. Tighten caps to prevent evaporation.

Soft Nutrient Agar (for bacteriophage)

Dehydrated nutrient broth	.8 gm
Agar	.7 gm
Distilled water	1000 ml

Sterilize in autoclave at 121° C for 20 minutes.

Spirit Blue Agar (Ex. 49)

This medium is used to detect lipase production by bacteria. Lipolytic bacteria cause the medium to change from pale lavender to deep blue.

Spirit blue agar (Difco)	.35 gm
Lipase reagent (Difco)	.35 ml
Distilled water	1000 ml

Dissolve the spirit blue agar in 1000 ml of water by boiling. Sterilize in autoclave for 15 minutes at 15 psi (121° C). Cool to 55° C and slowly add the 35 ml of lipase reagent, agitating to obtain even distribution. Dispense into sterile Petri plates.

Streptomycin Agar (Ex. 74)

To 1000 ml sterile liquid nutrient agar (50° C), aseptically add 100 mg of streptomycin sulfate. Pour directly into sterile Petri plates.

Top Agar (Ex. 76, Ames test)

Tubes containing 2 ml of top agar are made up just prior to using from bottles of top agar base and his/bio stock solution.

His/Bio Stock Solution

D-Biotin (F.W. 247.3)	.30.9 mg
L-Histidine · HCl (F.W. 191.7)	.24.0 mg
Distilled water	.250 ml

Dissolve by heating the water to the boiling point. This can be done in a microwave oven. Sterilize by filtration through 0.22 µm membrane filter, or autoclave for 20 minutes at 121° C. Store in a glass bottle at 4° C.

Top Agar Base

Agar	.6 gm
Sodium chloride (NaCl)	.5 gm
Distilled water	1000 ml

The agar may be dissolved in a steam bath or microwave oven, or by autoclaving briefly. Mix thoroughly and transfer 100-ml aliquots to 250-ml glass bottles with screw caps. Autoclave for 20 minutes with loosened caps. Slow exhaust. Cool the agar and tighten caps.

Just before using, add 10 ml of the his/bio stock solution to bottle of 100 ml of liquefied top agar base (45° C). After thoroughly mixing, distribute, aseptically, 2 ml of this mixture to sterile tubes (13 mm × 100 mm). Hold tubes at 45° C until used.

Tryptone Agar

Tryptone	.10 gm
Agar	.15 gm
Distilled water	1000 ml

Tryptone Broth

Tryptone	.10 gm
Distilled water	1000 ml

Tryptone Yeast Extract Agar

Tryptone	.10 gm
Yeast extract	.5 gm
Dipotassium phosphate	.3 gm
Sucrose	.50 gm
Agar	.15 gm
Water	1000 ml
pH 7.4	

D

Appendix Identification Charts

Appendix D • Identification Charts

Chart I Interpretation of Test Results of API 20E System

		Interpretation of Reactions		
Tube		Positive	Negative	Comments
ONPG		Yellow	Colorless	(1) Any shade of yellow is a positive reaction. (2) VP tube, before the addition of reagents, can be used as a negative control.
ADH	Incubation 18–24 h	Red or Orange	Yellow	Orange reactions occurring at 36–48 hours should be interpreted as negative.
	36–48 h	Red	Yellow or Orange	
LDC	18–24 h	Red or Orange	Yellow	Any shade of orange within 18–24 hours is a positive reaction. At 36–48 hours, orange decarboxylase reactions should be interpreted as negative.
	36–48 h	Red	Yellow or Orange	
ODC	18–24 h	Red or Orange	Yellow	Orange reactions occurring at 36–48 hours should be interpreted as negative.
	36–48 h	Red	Yellow or Orange	
CIT		Turquoise or Dark Blue	Light Green or Yellow	(1) Both the tube and cupule should be filled. (2) Reaction is read in the aerobic (cupule) area.
H ₂ S		Black Deposit	No Black Deposit	(1) H ₂ S production may range from a heavy black deposit to a very thin black line around the tube bottom. Carefully examine the bottom of the tube before considering the reaction negative. (2) A "browning" of the medium is a negative reaction unless a black deposit is present. "Browning" occurs with TDA-positive organisms.
URE	18–24 h	Red or Orange	Yellow	A method of lower sensitivity has been chosen. <i>Klebsiella</i> , <i>Proteus</i> , and <i>Yersinia</i> routinely give positive reactions.
	36–48 h	Red	Yellow or Orange	
TDA	Add 1 drop 10% ferric chloride			(1) Immediate reaction. (2) Indole-positive organisms may produce a golden orange color due to indole production. This is a negative reaction.
		Brown-Red	Yellow	
IND	Add 1 drop Kovacs' reagent			(1) The reaction should be read within 2 minutes after the addition of the Kovacs' reagent and the results recorded. (2) After several minutes, the HCl present in Kovacs' reagent may react with the plastic of the cupule resulting in a change from a negative (yellow) color to a brownish-red. This is a negative reaction.
		Red Ring	Yellow	
VP	Add 1 drop of 40% potassium hydroxide, then 1 drop of 6% alpha-naphthol.			(1) Wait 10 minutes before considering the reaction negative. (2) A pale pink color (after 10 min) should be interpreted as negative. A pale pink color appears immediately after the addition of reagents but turns dark pink or red after 10 min. should be interpreted as positive.
		Red	Colorless	
				Motility may be observed by hanging drop or wet mount preparation.
GEL		Diffusion of the pigment	No diffusion	(1) The solid gelatin particles may spread throughout the tube after inoculation. Unless diffusion occurs, the reaction is negative. (2) Any degree of diffusion is a positive reaction.
GLU		Yellow or Gray	Blue or Blue-Green	Comments for all Carbohydrates Fermentation (Enterobacteriaceae, <i>Aeromonas</i> , <i>Vibrio</i>) (1) Fermentation of the carbohydrates begins in the most anaerobic portion (bottom) of the tube. Therefore, these reactions should be read from the bottom of the tube to the top. (2) A yellow color at the bottom of the tube only indicates a weak or delayed positive reaction. Oxidation (Other Gram-negatives) (1) Oxidative utilization of the carbohydrates begins in the most aerobic portion (top) of the tube. Therefore, these reactions should be read from the top to the bottom of the tube. (2) A yellow color in the upper portion of the tube and a blue in the bottom of the tube indicates oxidative utilization of the sugar. This reaction should be considered positive only for non-Enterobacteriaceae gram-negative rods. This is a negative reaction for fermentative organisms such as Enterobacteriaceae.
MAN INO SOR RHA SAC MEL AMY ARA		Yellow	Blue or Blue-Green	
GLU	After reading GLU reaction, add 2 drops 0.8% sulfanilic acid and 2 drops 0.5% N, N-dimethylalpha-naphthylamine			
Nitrate Reduction	NO ₂ N ₂ gas	Red Bubbles; Yellow after reagents and zinc	Yellow Orange after reagents and zinc	
MAN INO SOR Catalase	After reading carbohydrate reaction, add 1 drop 1.5% H ₂ O ₂			
		Bubbles	No bubbles	(1) Bubbles may take 1–2 minutes to appear. (2) Best results will be obtained if the test is run in tubes which have no gas from fermentation.

Courtesy of Analytab Products, Plainview, N.Y.

Identification Charts • **Appendix D**

Chart II Symbol Interpretation of API 20E System

Tube	Chemical/Physical Principles	Components		Ref.
		Reactive Ingredients	Quantity	
ONPG	Hydrolysis of ONPG by beta-galactosidase releases yellow orthonitrophenol from the colorless ONPG; ITPG (isopropylthiogalactopyranoside) is used as inducer.	ONPG ITPG	0.2 mg 8.0 µg	12 13 14
ADH	Arginine dihydrolase transforms arginine into ornithine, ammonia, and carbon dioxide. This causes a pH rise in the acid-buffered system and a change in the indicator from yellow to red.	Arginine	2.0 mg	15
LDC	Lysine decarboxylase transforms lysine into a basic primary amine, cadaverine. This amine causes a pH rise in the acid-buffered system and a change in the indicator from yellow to red.	Lysine	2.0 mg	15
ODC	Ornithine decarboxylase transforms ornithine into a basic primary amine, putrescine. This amine causes a pH rise in the acid-buffered system and a change in the indicator from yellow to red.	Ornithine	2.0 mg	15
CIT	Citrate is the sole carbon source. Citrate utilization results in a pH rise and a change in the indicator from green to blue.	Sodium Citrate	0.8 mg	21
H ₂ S	Hydrogen sulfide is produced from thiosulfate. The hydrogen sulfide reacts with iron salts to produce a black precipitate.	Sodium Thiosulfate	80.0 µg	6
URE	Urease releases ammonia from urea; ammonia causes the pH to rise and changes the indicator from yellow to red.	Urea	0.8 mg	7
TDA	Tryptophane deaminase forms indolepyruvic acid from tryptophane. Indolepyruvic acid produces a brownish-red color in the presence of ferric chloride.	Tryptophane	0.4 mg	22
IND	Metabolism of tryptophane results in the formation of indole. Kovacs' reagent forms a colored complex (pink to red) with indole.	Tryptophane	0.2 mg	10
VP	Acetoin, an intermediary glucose metabolite, is produced from sodium pyruvate and indicated by the formation of a colored complex. Conventional VP tests may take up to 4 days, but by using sodium pyruvate, API has shortened the required test time. Creatine intensifies the color when tests are positive.	Sodium Pyruvate Creatine	2.0 mg 0.9 mg	3
GEL	Liquefaction of gelatin by proteolytic enzymes releases a black pigment which diffuses throughout the tube.	Kohn Charcoal Gelatin	0.6 mg	9
GLU MAN INO SOR RHA SAC MEL AMY ARA	Utilization of the carbohydrate results in acid formation and a consequent pH drop. The indicator changes from blue to yellow.	Glucose Mannitol Inositol Sorbitol Rhamnose Sucrose Mellibiose Amygdalin (l +) Arabinose	2.0 mg 2.0 mg 2.0 mg 2.0 mg 2.0 mg 2.0 mg 2.0 mg 2.0 mg 2.0 mg	5 6 12
GLU Nitrate Reduction	Nitrites form a red complex with sulfanilic acid and N, N'-dimethyl-alpha-naphthylamine. In case of negative reaction, addition of zinc confirms the presence of unreduced nitrates by reducing them to nitrites (pink-orange color). If there is no color change after the addition of zinc, this is indicative of the complete reduction of nitrates through nitrites to nitrogen gas or to an anaerogenic amine.	Potassium Nitrate	80.0 µg	6
MAN INO SOR Catalase	Catalase releases oxygen gas from hydrogen peroxide.			24

Courtesy of Analytab Products, Plainview, N.Y.

Appendix D • Identification Charts

Chart III Characterization of Gram-Negative Rods—The API 20E System

ORGANISM	ONPG	ADH	LDC	ODC	CIT	H ₂ S	URE	TDA	IND	VP	GEL	GLU	MAN	INO	SOR	RHA	SAC	MEL	AMY	ARA	OXI
<i>Escherichia</i>																					
<i>E. coli</i>	98.2	1.0	90.2	67.3	0	1.0	0	0	85.0	0	0	100	98.4	0.1	95.5	84.5	41.1	88.4	0.1	95.0	0
<i>Shigella dysenteriae</i>	27.8	0	0	0	0	0	0	0	33.0	0	0	100	0.1	0	0	22.2	0	61.1	0	16.7	0
<i>Sh. flexneri</i>	5.3	0	0	0	0	0	0	0	15.0	0	0	100	94.7	0	78.9	0	0	21.1	0	36.8	0
<i>Sh. boydii</i>	5.0	0	0	0	0	0	0	0	20.0	0	0	100	60.0	0	53.3	1.0	0	33.3	0	66.7	0
<i>Sh. sonnei</i>	96.7	0	0	80.0	0	0	0	0	0	0	0	100	99.0	0	39.9	80.0	0	50.0	0	96.7	0
<i>Edwardsiella tarda</i>	0	0	99.0	99.0	0	55.0	0	0	100	0	0	100	0	0	0	0	0	50.0	0	1.1	0
<i>Salmonella enteritidis</i>	1.9	1.0	89.2	95.4	15.4	76.9	0	0	3.1	0	0	100	98.7	4.6	95.2	95.4	4.6	96.9	0	94.5	0
<i>Sal. typhi</i>	0	0	90.0	0	0	0.1	0	0	0	0	0	100	99.0	0	99.0	1.8	0	100	0	27.0	0
<i>Sal. paratyphi A</i>	0	0	0	100	0	0.2	0	0	0	0	0	100	99.0	0	99.0	99.0	0	40.0	0	80.0	0
<i>Arizona-S. arizonae</i>	94.7	1.0	95.0	98.5	15.0	85.0	0	0	0	0	0	100	99.0	0	87.0	96.1	0	89.5	0	95.0	0
<i>Citrobacter freundii</i>	97.0	10.0	0	60.0	10.0	81.0	0	0	6.0	0	0	100	98.0	1.0	96.0	87.0	59.0	77.0	30.0	98.0	0
<i>C. diversus-Levinea</i>	97.0	10.0	0	90.0	10.0	0	0	0	91.0	0	0	100	97.0	14.5	88.0	99.0	51.0	47.0	34.0	99.0	0
<i>C. anatum</i>	97.0	10.0	0	95.0	10.0	0	0	0	99.0	0	0	100	97.0	0.1	93.0	99.0	29.4	53.0	80.0	93.8	0
<i>Klebsiella pneumoniae</i>	99.0	0	80.0	0	13.9	0	10.0	0	0	72.0	0	100	98.0	30.0	95.0	91.0	99.0	99.0	96.0	99.0	0
<i>K. oxytoca</i>	98.0	0	83.0	0	13.0	0	10.0	0	100	60.0	1.0	100	99.0	29.0	92.0	98.0	99.0	99.0	98.0	99.0	0
<i>K. ozaenae</i>	85.0	0	38.0	0	1.0	0	0	0	0	0	0	100	69.0	1.0	76.0	69.0	15.0	92.0	99.0	84.0	0
<i>K. rhinoscleromatis</i>	0	0	0	0	0	0	0	0	0	0	0	100	99.0	1.0	86.0	53.0	33.0	66.0	99.0	95.0	0
<i>Enterobacter aerogenes</i>	99.0	0	98.0	98.0	8.9	0	0	0	0	56.0	0	100	99.0	28.0	90.0	90.0	85.0	97.0	95.0	98.0	0
<i>Ent. cloacae</i>	97.0	51.9	0	65.0	9.0	0	0	0	0	80.0	0	100	99.0	1.0	92.0	90.0	98.0	92.0	65.0	95.0	0
<i>Ent. agglomerans</i>	90.0	0	0	0	5.4	0	0	0	50.0	20.0	0	100	99.0	1.0	80.0	60.0	60.0	70.0	70.0	95.0	0
<i>Ent. gergoviae</i>	99.0	0	61.0	99.0	8.2	0	75.0	0	0	75.0	0	100	99.0	1.0	8.3	99.0	99.0	99.0	99.0	99.0	0
<i>Ent. sakazakii</i>	97.0	51.6	0	59.0	8.6	0	0	0	4.0	85.0	0	100	99.0	4.0	8.5	90.0	95.0	90.0	76.0	95.0	0
<i>Serratia liquefaciens</i>	85.0	0	85.0	95.0	8.9	0	1.0	0	0	50.0	60.0	100	99.0	1.0	99.0	30.0	85.0	80.7	80.0	92.9	0
<i>Ser. marcescens</i>	83.0	0	88.0	94.0	8.0	0	1.0	0	0	58.0	72.0	100	96.0	1.0	97.0	2.0	98.0	37.0	72.0	18.0	0
<i>Ser. rubidaea</i>	96.0	0	60.5	0.1	8.2	0	0	0	0	70.0	75.6	100	99.0	10.0	75.0	13.4	99.0	82.6	96.0	85.8	0
<i>Ser. odorifera 1</i>	99.0	0	95.0	100	9.5	0	0	0	90.0	63.0	62.0	100	99.0	10.0	99.0	85.0	100	99.0	90.0	99.0	0
<i>Ser. odorifera 2</i>	99.0	0	92.0	0	9.1	0	0	0	90.0	80.0	78.0	100	99.0	10.0	99.0	95.0	0	99.0	85.4	99.0	0
<i>Hafnia alvei</i>	60.0	0	99.0	99.0	1.0	0	0	0	0	25.0	0	99.0	99.0	0	35.0	75.0	0	50.0	30.0	95.0	0
<i>Klebsiella</i>																					

Identification Charts • Appendix D

Chart III (continued)

ORGANISM	ONPG	ADH	LDC	ODC	CIT	H ₂ S	URE	TDA	IND	VP	GEL	GLU	MAN	INO	SOR	RHA	SAC	MEL	AMY	ARA	OXI
<i>Proteus vulgaris</i>	0.5	0	0	0	4.1	75.3	91.0	95.0	75.3	0	75.3	100	0	0.1	0	0	83.0	1.0	20.0	4.0	0
<i>Prot. mirabilis</i>	1.0	0	0	90.0	5.8	66.0	97.0	90.0	1.0	0	93.0	100	0	0	0	1.0	9.6	10.0	1.0	27.0	0
<i>Providencia alcalifaciens</i>	0	0	0	0	9.8	0	0	95.0	94.0	0	0	100	0	0	0	0	0	0	0	25.0	0
<i>Prov. stuartii</i>	1.0	0	0	0	8.5	0	0	95.0	86.0	0	0	100	0	8.0	0	0.8	3.7	34.0	0	30.0	0
<i>Prov. stuartii</i> URE +	1.0	0	0	0	6.9	0	99.0	99.0	95.0	0	0	100	15.0	5.0	0	0.5	65.0	20.0	0	20.0	0
<i>Prov. rettgeri</i>	1.0	0	0	0	7.1	0	80.0	95.0	90.0	0	0	100	85.0	1.0	30.0	40.0	5.0	0	40.0	10.0	0
<i>Morganella morganii</i>	1.0	0	0	87.0	0.2	0	78.0	92.0	92.0	0	0	98.0	0	0	0	0	0	0	0	1.0	0
<i>Yersinia enterocolitica</i>	81.0	0	0	36.0	0	0	59.0	0	54.0	0.4	0	100	99.0	1.0	95.0	9.0	78.0	40.4	31.0	76.6	0
<i>Y. pseudotuberculosis</i>	80.0	0	0	0	0	0	88.0	0	0	0	0	100	94.0	0	76.0	58.0	0	5.0	0	52.0	0
<i>Y. pestis</i>	93.0	0	0	0	0	0	0	0	0	1.0	0	93.0	87.0	0	56.0	0	0	0.6	25.0	87.0	0
API Group 1	99.0	0	58.8	99.0	9.2	0	0	0	99.0	0	0	100	99.0	0	75.4	82.4	82.4	94.1	97.0	94.1	0
API Group 2	99.0	2.0	7.3	0	0	0	0	0	0	0	0	100	99.0	0	2.3	30.8	5.6	90.0	38.5	92.3	0
<i>Pseudomonas maltophilia</i>	62.0	0	5.0	0	7.6	0	0	0	0	0	50.0	0.5	0	0	0	0	0	0	0	22.0	4.8
<i>Ps. cepacia</i>	61.0	0	5.0	5.0	7.5	0	0	0	0	1.0	46.0	33.0	1.0	0	1.0	0	7.0	0	1.0	1.0	90.7
<i>Ps. paucimobilis</i>	40.0	0	0	0	1.0	0	0	0	0	0	0	0.5	0	0	0	0	0.5	0	0	0.5	50.0
<i>A. calco</i> var. <i>antratus</i>	0	0	0	0	2.8	0	0	0	0	1.0	0.1	85.0	0	0	0	0	0.1	77.0	0	60.0	0
<i>A. calco</i> var. <i>lwoffi</i>	0	0	0	0	0	0	0	0	0	0.1	0	0	0	0	0	0	0	0	0	0	0
CDC Group VE-1	90.0	1.0	0	0	7.7	0	0	0	0	1.0	1.3	33.0	0	1.0	0	1.0	0.1	1.0	1.0	16.0	0
CDC Group VE-2	0	0	0	0	7.9	0	0	0	0	1.0	0.1	4.5	0	1.0	0	0	0	1.0	0	5.0	0

Courtesy of Analytab Products, Plainview, N.Y.

Appendix D • Identification Charts

Chart IV Characterization of Enterobacteriaceae—The Enterotube II System

Groups		REACTIONS	GLUCOSE	GAS PRODUCTION	LYSINE	ORNITHINE	H ₂ S	INDOLE	ADONITOL	LACTOSE	ARABINOSE	SORBITOL	VOGES-PROSKAUER	DULCITOL	PHENYLALANINE DEAMINASE	UREA	CITRATE
ESCHERICHIEAE	<i>Escherichia</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>Shigella</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
EDWARDSIELLEAE	<i>Edwardsiella</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>Salmonella</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
SALMONELLEAE	<i>Anzonia</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>freundii</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
CITROBACTER	<i>amalonaticus</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>diversus</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
PROTEUS	<i>vulgaris</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>mirabilis</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
MORGANELLA	<i>morganii</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>alcalifaciens</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
PROVIDENCIA	<i>stuartii</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>rettgeri</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
ENTEROBACTER	<i>cloacae</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>sakazakii</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
HAFNIA	<i>alvei</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>marcescens</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
SERRATIA	<i>liquefaciens</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>rubidaea</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
KLEBSIELLEAE	<i>pneumoniae</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>oxytoca</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
KLEBSIELLA	<i>ozaenae</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>rhinoschleromatis</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
YERSINIAE	<i>enterocolitica</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
	<i>pseudotuberculosis</i>	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+

Courtesy of Roche Diagnostics, Nutley, N.J.

E. *S. enteritidis* bioserotype Paratyphi A and some rare biotypes may be H₂S negative.F. *S. typhi*, *S. enteritidis* bioserotype Paratyphi A and some rare biotypes are citrate-negative and *S. cholerae-suis* is usually delayed positive.G. The amount of gas produced by *Serratia*, *Proteus* and *Providencia alcalifaciens* is slight; therefore, gas production may not be evident in the ENTEROTUBE II.H. *S. enteritidis* bioserotype Paratyphi A is negative for lysine decarboxylase.I. *S. typhi* and *S. gallinarum* are ornithine decarboxylase-negative.J. The *Alcaligenes*-Dispar (A-D) group is included as a biotype of *E. coli*. Members of the A-D group are generally anaerogenic, non-motile and do not ferment lactose.

K. An occasional strain may produce hydrogen sulfide.

L. An occasional strain may appear to utilize citrate.

Identification Charts • **Appendix D**

Chart V Reaction Interpretations for API Staph-Ident

MICROCUPULE		INTERPRETATION OF REACTIONS		
NO.	SUBSTRATE	POSITIVE	NEGATIVE	COMMENTS AND REFERENCES
1	PHS	Yellow	Clear or straw-colored	A positive result should be recorded only if significant color development has occurred.(3)
2	URE	Purple to Red-Orange	Yellow or Yellow-Orange	Phenol red has been added to the urea formulation to allow detection of alkaline end products resulting from urea utilization.(1)
3	GLS	Yellow	Clear or straw-colored	A positive result should be recorded only if significant color development has occurred.
4 5 6 7	MNE MAN TRE SAL	Yellow or Yellow-Orange	Red or Orange	Cresol red has been added to each carbohydrate to allow detection of acid production if the respective carbohydrates are utilized. (1,7)
8	GLC	Yellow	Clear or straw-colored	A positive result should be recorded only if significant color development has occurred.
9	ARG	Purple to Red-Orange	Yellow or Yellow-Orange	Phenol red has been added to the arginine formulation to allow detection of alkaline end products resulting from arginine utilization.(1)
10	NGP	Add 1-2 drops of STAPH-IDENT REAGENT Plum-Purple (Mauve) Yellow or colorless		Color development will begin within 30 seconds of reagent addition. (1,5)

Courtesy of Analytab Products, Plainview, N.Y.

Abbreviation

PHS
URE
GLS
MNE
MAN
TRE
SAL
GLC
ARG
NGP

Test

Phosphatase
Urea utilization
 β -Glucosidase
Mannose utilization
Mannitol utilization
Trehalose utilization
Salicin utilization
 β -Glucuronidase
Arginine utilization
 β -Galactosidase

Appendix D • Identification Charts

Chart VI Biochemistry of API Staph-Ident Tests

MICROCUPULE		CHEMICAL/PHYSICAL PRINCIPLES	REACTIVE INGREDIENTS	QUANTITY
NO.	SUBSTRATE			
1	PHS	Hydrolysis of p-nitrophenyl-phosphate, disodium salt, by alkaline phosphatase releases yellow paranitrophenol from the colorless substrate.	p-nitrophenyl-phosphate, disodium salt	0.2%
2	URE	Urease releases ammonia from urea; ammonia causes the pH to rise and changes the indicator from yellow to red.	Urea	1.6%
3	GLS	Hydrolysis of p-nitrophenyl- β -D-glucopyranoside by β -glucosidase releases yellow para-nitrophenol from the colorless substrate.	p-nitrophenyl- β -D-glucopyranoside	0.2%
4 5 6 7	MNE MAN TRE SAL	Utilization of carbohydrate results in acid formation and a consequent pH drop. The indicator changes from red to yellow.	Mannose Mannitol Trehalose Salicin	1.0% 1.0% 1.0% 1.0%
8	GLC	Hydrolysis of p-nitrophenyl- β -D-glucuronide by β -glucuronidase releases yellow para-nitrophenol from the colorless substrate.	p-nitrophenyl- β -D-glucuronide	0.2%
9	ARG	Utilization of arginine produces alkaline end products which change the indicator from yellow to red.	Arginine	1.6%
10	NGP	Hydrolysis of 2-naphthol- β -D-galactopyranoside by β -galactosidase releases free β -naphthol which complexes with STAPH-IDENT REAGENT to produce a plum-purple (mauve) color.	2-naphthol- β -D-galactopyranoside	0.3%

Courtesy of Analytab Products, Plainview, N.Y.

Identification Charts • **Appendix D**

Chart VII API Staph-Ident Profile Register*

Profile	Identification	Profile	Identification
0 040	STAPH CAPITIS	4 700	STAPH AUREUS
0 060	STAPH HAEMOLYTICUS		STAPH SCIURI
0 100	STAPH CAPITIS	4 710	STAPH SCIURI
0 140	STAPH CAPITIS	5 040	STAPH EPIDERMIDIS
0 200	STAPH COHNII	5 200	STAPH SCIURI
0 240	STAPH CAPITIS	5 210	STAPH SCIURI
0 300	STAPH CAPITIS	5 300	STAPH AUREUS
0 340	STAPH CAPITIS		STAPH SCIURI
0 440	STAPH HAEMOLYTICUS		STAPH SCIURI
0 460	STAPH HAEMOLYTICUS	5 310	STAPH SCIURI
0 600	STAPH COHNII	5 600	STAPH SCIURI
0 620	STAPH HAEMOLYTICUS	5 610	STAPH SCIURI
0 640	STAPH HAEMOLYTICUS	5 700	STAPH AUREUS
0 660	STAPH HAEMOLYTICUS		STAPH SCIURI
1 000	STAPH EPIDERMIDIS	5 710	STAPH SCIURI
1 040	STAPH EPIDERMIDIS	5 740	STAPH AUREUS
1 300	STAPH AUREUS	6 001	STAPH XYLOSUS
1 540	STAPH HYICUS (An)		STAPH SAPROPHYTICUS
1 560	STAPH HYICUS (An)	6 011	STAPH XYLOSUS
2 000	STAPH SAPROPHYTICUS	6 021	STAPH XYLOSUS
	STAPH HOMINIS	6 101	STAPH XYLOSUS
	NOVO R	6 121	STAPH XYLOSUS
2 001	STAPH SAPROPHYTICUS	6 221	STAPH XYLOSUS
	NOVO S	6 300	STAPH AUREUS
2 040	STAPH SAPROPHYTICUS	6 301	STAPH XYLOSUS
	NOVO S	6 311	STAPH XYLOSUS
2 041	STAPH SIMULANS	6 321	STAPH XYLOSUS
2 061	STAPH SIMULANS	6 340	STAPH AUREUS
2 141	STAPH SIMULANS		STAPH WARNERI
2 161	STAPH SIMULANS	6 400	STAPH WARNERI
2 201	STAPH SAPROPHYTICUS	6 401	STAPH XYLOSUS
2 241	STAPH SIMULANS		STAPH SAPROPHYTICUS
2 261	STAPH SIMULANS	6 421	STAPH XYLOSUS
2 341	STAPH SIMULANS	6 460	STAPH WARNERI
2 361	STAPH SIMULANS	6 501	STAPH XYLOSUS
2 400	STAPH HOMINIS	6 521	STAPH XYLOSUS
	NOVO S	6 600	STAPH WARNERI
	NOVO R	6 601	STAPH SAPROPHYTICUS
2 401	STAPH SAPROPHYTICUS		STAPH XYLOSUS
2 421	STAPH SIMULANS	6 611	STAPH XYLOSUS
2 441	STAPH SIMULANS	6 621	STAPH XYLOSUS
2 461	STAPH SIMULANS	6 700	STAPH AUREUS
2 541	STAPH SIMULANS	6 701	STAPH XYLOSUS
2 561	STAPH SIMULANS	6 721	STAPH XYLOSUS
2 601	STAPH SAPROPHYTICUS	6 731	STAPH XYLOSUS
2 611	STAPH SAPROPHYTICUS	7 000	STAPH EPIDERMIDIS
2 661	STAPH SIMULANS	7 021	STAPH XYLOSUS
2 721	STAPH COHNII (SSP1)	7 040	STAPH EPIDERMIDIS
2 741	STAPH SIMULANS	7 141	STAPH INTERMEDIUS (An)
2 761	STAPH SIMULANS	7 300	STAPH AUREUS
3 000	STAPH EPIDERMIDIS	7 321	STAPH XYLOSUS
3 040	STAPH EPIDERMIDIS	7 340	STAPH AUREUS
3 140	STAPH EPIDERMIDIS	7 401	STAPH XYLOSUS
3 540	STAPH HYICUS (An)	7 421	STAPH XYLOSUS
3 541	STAPH INTERMEDIUS (An)	7 501	STAPH INTERMEDIUS (An)
3 560	STAPH HYICUS (An)		STAPH XYLOSUS
3 601	STAPH SIMULANS		STAPH XYLOSUS
	NOVO S	7 521	STAPH XYLOSUS
	NOVO R	7 541	STAPH INTERMEDIUS (An)
4 060	STAPH HAEMOLYTICUS	7 560	STAPH HYICUS (An)
4 210	STAPH SCIURI	7 601	STAPH XYLOSUS
4 310	STAPH SCIURI	7 621	STAPH XYLOSUS
4 420	STAPH HAEMOLYTICUS	7 631	STAPH XYLOSUS
4 440	STAPH HAEMOLYTICUS	7 700	STAPH AUREUS
4 460	STAPH HAEMOLYTICUS	7 701	STAPH XYLOSUS
4 610	STAPH SCIURI	7 721	STAPH XYLOSUS
4 620	STAPH HAEMOLYTICUS	7 740	STAPH AUREUS
4 660	STAPH HAEMOLYTICUS		

*Date of Publication: March, 1984

Courtesy of Analytab Products, Plainville, N.Y.

E

Appendix

The Streptococci: Classification, Habitat, Pathology, and Biochemical Characteristics

To fully understand the characteristics of the various species of medically important streptococci, this appendix has been included as an adjunct to Exercise 79. The table of streptococcal characteristics on this page is the same one that is shown on page 267 of Exercise 79. It is also the basis for much of the discussion that follows.

The first system that was used for grouping the streptococci was based on the type of hemolysis and was proposed by J. H. Brown in 1919. In 1933, R. C. Lancefield proposed that these bacteria be separated into groups A, B, C, etc., on the basis of precipitation-type serological testing. Both hemolysis and serological typing still play predominant roles today in our

classification system. Note below that the Lancefield groups are categorized with respect to the type of hemolysis that is produced on blood agar.

Beta Hemolytic Groups

Using a streak-stab technique, a blood agar plate is incubated aerobically at 37° C for 24 hours. Isolates that have colonies surrounded by clear zones completely free of red blood cells are characterized as being *beta hemolytic*. Three serological groups of streptococci fall in this category: groups A, B, and C; a few species in group D are also beta hemolytic.

Table 1 Physiological Tests for Streptococcal Differentiation

GROUP	Type of Hemolysis	Bacitracin Susceptibility	CAMP Reaction or Hippurate Hydrolysis	SXT Sensitivity	Bile Esculin Hydrolysis	Tolerance to 6.5% NaCl	Optochin Susceptibility	Bile Solubility	
Group A <i>S. pyogenes</i>	beta	+	—	R	—	—	—	—	
Group B <i>S. agalactiae</i>	beta	—*	+	R	—	±	—	—	
Group C <i>S. equi</i> <i>S. equisimilis</i> <i>S. zooepidemicus</i>	beta	—*	—	S	—	—	—	—	
Group D** (enterococci) <i>S. faecalis</i> <i>S. faecium</i> etc.	alpha beta none	—	—	R	+	+	—	—	
Group D** (nonenterococci) <i>S. bovis</i> etc.	alpha none	—	—	R/S	+	—	—	—	
Viridans <i>S. mitis</i> <i>S. salivarius</i> <i>S. mutans</i> etc.	alpha none	—*	—*	S	—	—	—	—	
Pneumococci <i>S. pneumoniae</i>	alpha	±	—		—	—	+	+	

*Exceptions occur occasionally **See comments on pp. 457 and 458 concerning correct genus.

Note: R = resistant; S = sensitive; blank = not significant

Appendix E • The Streptococci: Classification, Habitat, Pathology, and Biochemical Characteristics**Group A Streptococci**

This group is represented by only one species: *Streptococcus pyogenes*. Approximately 25% of all upper respiratory infections (URIs) are caused by this species; another 10% of URIs are caused by other streptococci; most of the remainder (65%) are caused by viruses. Since no unique clinical symptoms can be used to differentiate viral from streptococcal URIs, and since successful treatment relies on proper identification, it becomes mandatory that throat cultures be taken in an attempt to prove the presence or absence of streptococci. It should be added that if streptococcal URIs are improperly treated, serious sequelae such as pneumonia, acute endocarditis, rheumatic fever, and glomerulonephritis can result.

S. pyogenes is the only beta hemolytic streptococcus that is primarily of *human origin*. Although the pharynx is the most likely place to find this species, it may be isolated from the skin and rectum. Asymptomatic pharyngeal and anal carriers are not uncommon. Outbreaks of postoperative streptococcal infections have been traced to both pharyngeal and anal carriers among hospital personnel.

These coccoidal bacteria (0.6–1.0 μm diameter) occur as pairs and as short to moderate-length chains in clinical specimens; in broth cultures, the chains are often longer.

When grown on blood agar, the colonies are small (0.5 mm dia.), transparent to opaque, and domed; they have a smooth or semimatte surface and an entire edge; complete hemolysis (beta-type) occurs around each colony, usually two to four times the diameter of the colony.

S. pyogenes produces two hemolysins: streptolysin S and streptolysin O. The beta-type hemolysis on blood agar is due to the complete destruction of red blood cells by the streptolysin S.

There is no group of physiological tests that can be used with *absolute* certainty to differentiate *S. pyogenes* from other streptococci; however, if an isolate is beta hemolytic and sensitive to bacitracin, one can be 95% certain that the isolate is *S. pyogenes*. The characteristics of this organism are the first ones tabulated in table I on the previous page.

Group B Streptococci

The only recognized species of this group is *S. agalactiae*. Although this organism is frequently found in milk and associated with *mastitis in cattle*, the list of human infections caused by it is as long as the one for *S. pyogenes*: abscesses, acute endocarditis, impetigo, meningitis, neonatal sepsis, and pneumonia are just a few. Like *S. pyogenes*, this pathogen may also be found in the pharynx, skin, and rectum; how-

ever, it is more likely to be found in the genital and intestinal tracts of healthy adults and infants. It is not unusual to find the organism in vaginal cultures of third-trimester pregnant women.

Cells are spherical to ovoid (0.6–1.2 μm dia) and occur in chains of seldom less than four cells; long chains are frequently present. Characteristically, the chains appear to be composed of paired cocci.

Colonies of *S. agalactiae* on blood agar often produce double zone hemolysis. After 24 hours incubation colonies exhibit zones of beta hemolysis. After cooling, a second ring of hemolysis forms which is separated from the first by a ring of red blood cells.

Reference to table I emphasizes the significant characteristics of *S. agalactiae*. Note that this organism gives a positive CAMP reaction, hydrolyzes hippurate, and is not (usually) sensitive to bacitracin. It is also resistant to SXT. Presumptive identification of this species relies heavily on a positive CAMP test or hippurate hydrolysis, even if beta hemolysis is not clearly demonstrated.

Group C Streptococci

Three species fall in this group: *S. equisimilis*, *S. equi*, and *S. zooepidemicus*. Although all of these species may cause human infections, the diseases are not usually as grave as those caused by groups A and B. Some group C species have been isolated from impetiginous lesions, abscesses, sputum, and the pharynx. There is no evidence that they are associated with acute glomerulonephritis, rheumatic fever, or even pharyngitis.

Presumptive differentiation of this group from *S. pyogenes* and *S. agalactiae* is based primarily on (1) resistance to bacitracin, (2) inability to hydrolyze hippurate or bile esculin, and (3) a negative CAMP test. There are other groups that have some of these same characteristics, but they will not be studied here. Tables 12.16 and 12.17 on page 1049 of *Bergey's Manual*, vol. 2, provide information about these other groups.

Alpha Hemolytic Groups

Streptococcal isolates that have colonies with zones of incomplete lysis around them are said to be **alpha hemolytic**. These zones are often greenish; sometimes they are confused with beta hemolysis. *The only way to be certain that such zones are not beta hemolytic is to examine the zones under 60 $\times$ microscopic magnification.* Figure 79.4, page 265, illustrates the differences between alpha and beta hemolysis. If some red blood cells are seen in the zone, the isolate is classified as being alpha hemolytic.

The Streptococci: Classification, Habitat, Pathology, and Biochemical Characteristics • Appendix E

The grouping of streptococci on the basis of alpha hemolysis is not as clear-cut as it is for beta hemolytic groups. Note in table I that the bottom four groups that have alpha hemolytic types may also have beta hemolytic or nonhemolytic strains. Thus, we see that hemolysis in these four groups can be a misleading characteristic in identification.

Alpha hemolytic isolates from the pharynx are usually *S. pneumoniae*, viridans streptococci, or group D. Our primary concern here in this experiment is to identify isolates of *S. pneumoniae*. To accomplish this goal, it will be necessary to differentiate any alpha hemolytic isolate from group D and viridans streptococci.

Streptococcus pneumoniae
(Pneumococcus)

This organism is the most frequent cause of bacterial pneumonia, a disease that has a high mortality rate among the aged and debilitated. It is also frequently implicated in conjunctivitis, otitis media, pericarditis, subacute endocarditis, meningitis, septicemia, empyema, and peritonitis. Thirty to 70% of normal individuals carry this organism in the pharynx.

Spherical or ovoid, these cells (0.5–1.25 μm dia) occur typically as pairs, sometimes singly, often in short chains. Distal ends of the cells are pointed or lancet-shaped and are heavily encapsulated with polysaccharide on primary isolation.

Colonies on blood agar are small, mucoid, opalescent, and flattened with entire edges surrounded by a zone of greenish discoloration (alpha hemolysis). In contrast, the viridans streptococcal colonies are smaller, gray to whitish gray, and opaque with entire edges.

Presumptive identification of *S. pneumoniae* can be made with the optochin and bile solubility tests. On the optochin test, the pneumococci exhibit sensitivity to ethylhydrocupreine (optochin). With the bile solubility test, pneumococci are dissolved in bile (2% sodium desoxycholate). Table I reveals that except for bacitracin susceptibility ($\pm$), *S. pneumoniae* is negative on all other tests used for differentiation of streptococci.

Viridans Group

Streptococci that fall in this group are primarily alpha hemolytic; some are nonhemolytic. Approximately 10 species are included in this group. All of them are highly adapted parasites of the upper respiratory tract. Although usually regarded as having low pathogenicity, they are opportunistic and sometimes cause serious infections. Two species (*S. mutans* and *S. sanguis*) are thought to be the primary cause of dental caries,

since they have the ability to form dental plaque. Viridans streptococci are implicated more often than any other bacteria in subacute bacterial endocarditis.

When it comes to differentiation of bacteria of this group from the pneumococci and enterococci, we will use the optochin, bile solubility, and salt-tolerance tests. See table I.

Group D Streptococci (Enterococci)

Members of this group are, currently, considered by most taxonomists to belong to the genus *Enterococcus*. During the preparation of volume 2 of *Bergey's Manual* Schleifer and Kilper-Balz presented conclusive evidence that *S. faecalis*, *S. faecium*, and *S. bovis* were so distantly related to the other groups of streptococci that they should be transferred to another genus. Since the term *Enterococcus* had been previously suggested by others, Schleifer and Kilper-Balz recommended that this be the name of a new genus to include all of the Group D streptococci, nonenterococci included. The fact that these papers came too late for *Bergey's Manual* to include this new genus caused the genus *Streptococcus* to be retained. To avoid confusion in our use of *Bergey's Manual*, we have retained the same terminology used in *Bergey's Manual*.

The enterococci of serological group D may be alpha hemolytic, beta hemolytic, or nonhemolytic. The principal species of this enterococcal group are *S. faecalis*, *S. faecium*, *S. durans*, and *S. avium*.

Subacute endocarditis, pyelonephritis, urinary tract infections, meningitis, and biliary infections are caused by these organisms. All five of these species have been isolated from the intestinal tract. Approximately 20% of subacute bacterial endocarditis and 10% of urinary tract infections are caused by members of this group. Differentiation of this group from other streptococci in systemic infections is mandatory because *S. faecalis*, *S. faecium*, and *S. durans* are resistant to penicillin and require combined antibiotic therapy.

Since *S. faecalis* can be isolated from many food products (not connected with fecal contamination), it can be a transient in the pharynx and show up as an isolate in throat cultures. Morphologically, the cells are ovoid (0.5–1.0 μm dia) occurring as pairs in short chains. Hemolytic reactions of *S. faecalis* on blood agar will vary with the type of blood used in the medium. Some strains produce beta hemolysis on agar with horse, human, and rabbit blood; on sheep blood agar the colonies will always exhibit alpha hemolysis. Other streptococci are consistently either beta, alpha, or nonhemolytic.

Cells of *S. faecium* are morphologically similar to *S. faecalis* except that motile strains are often encoun-

Appendix E • The Streptococci: Classification, Habitat, Pathology, and Biochemical Characteristics

tered. A strong alpha-type hemolysis is usually seen around colonies of *S. faecium* on blood agar.

Although presumptive differentiation of group D enterococcal streptococci from groups A, B, and C is not too difficult with physiological tests, it is more laborious to differentiate the individual species within group D. As indicated in table I, the enterococci (1) hydrolyze bile esculin, (2) are CAMP negative, and (3) grow well in 6.5% NaCl broth.

Differentiation of the five species within this group involves nine or ten physiological tests.

Group D Streptococci (Nonenterococci)

The only medically significant nonenterococcal species of group D is *S. bovis*. This organism is found in the intestinal tract of humans as well as in cows, sheep, and other ruminants. It can cause meningitis, subacute endocarditis, and urinary tract infections. On blood agar, the organism is usually alpha hemolytic; occasionally, it is nonhemolytic. The best way to differentiate it from the group D enterococci is to test its tolerance to 6.5% NaCl. Note in table I that *S. bovis* will not grow in this medium, but all enterococci will.

F

Appendix *Identibacter Interactus*

As stated in Exercise 51, *Identibacter interactus* is a computer program designed to assist students in identifying unknown bacterial cultures. This CD-ROM program, which is distributed by WCB/McGraw-Hill Co. in Dubuque, is a powerful program that includes more than 50 tests to run on assigned bacterial unknowns. The organism data base includes about 60 species of chemoheterotrophic bacteria.

To run this program, you will select each test from pull-down menus. A color image of each test result will be displayed on the computer screen, and you must be able to correctly interpret the test result that

is shown. Once you have tabulated sufficient information, you can identify your unknown by typing in the name of the organism. An audit trail of your choices can be saved to disk which can be evaluated by your instructor.

Before you attempt to use this program, read over the following pages of this Appendix. These twelve pages are the first portion of a 59 page instructional manual that can be accessed from the CD-ROM. This information will explain more in detail how the program functions. A full copy of the manual should be available to you in the laboratory.

Index

- Acetobacter*, 241
acid-fast staining, 69
Actinomycetes, isolation of, 203–5
agglutination tests
 Epstein-Barr virus, 283
 heterophile antibody, 283
 S. aureus, 281
 Widal, 285
Alcaligenes, characteristics of, 180, 270
alcohol fermentation, 241
Algae, Subkingdom, 26
alpha hemolysis, 264
alpha toxin, 258
Amastigomycota, 49
ammonification, 212
amoeboid movement, 28
Anabaena, 34
anaerobe culture, 89
anaerobic phototrophic bacteria, 106
Annelida, 36
antagonism, microbial, 128
antibiotic production, in soil, 203
antibiotic testing, 145
antigens, heterophile, 283
antiseptics, evaluation of, 143
Apicomplexa, 28
API 20E system, 185
API Staph-Ident system, 198
Archaea, Domain, 25
Arthrobacter, characteristics of, 179
arthrospores, 49
Ascomycetes, 50
ascospores, 49
Aschelminthes, 35
aseptic technique, 39–45
Aspergillus, 52, 53
atomic weights, 425
autoclave steam pressure table, 429
autotrophs, 76
Azotobacter, 210
- Bacillus*, characteristics, 178
bacitracin susceptibility, 267
bacteria, definition, 46
Bacteria, Domain, 25
bacteriochlorophyll, 30, 46
bacteriophage, 111–24
Barritt's reagents, usage, 167
Basidiomycotina, 50
basidiospores, 4
basophils, 288, 289
Bergey's Manual, usage of, 177–81
beta hemolysis, 264
bile esculin hydrolysis, 268
bile solubility test, 269
blastocoonidia, 49
blastospore, 49
- blood agar usage, 260
blood cells
 differential WBC count, 288
 total WBC count, 292
 typing, 295
blood types, 296
Bradyrhizobium, 207, 211
Breed count, 231
burst size, phage, 120
butanediol fermentation, 167
- CAMP test, 262, 266
capsid, 112
capsular staining, 63
cardioid condenser, 10
caries susceptibility test, 299
carotene, 30
casein hydrolysis test, 172
catalase test, 168
Ceratium, 30, 31
chemoautotrophs, 77
Chlamydomonas, 28, 29
chlamydospores, 49
Chlorobiaceae, 106
Chlorobium, 107
Chlorogonium, 29
chlorophyll, 28
chloroplasts, 28
Chromatiaceae, 106
Chromatium, 107
Chrysophycophyta, 30
Ciliophora, 28
citrate utilization test, 175
Citrobacter, 270
cladocera, 36, 37
Cladosporium, 52, 53
Clostridium, characteristics, 178
coagulase test, 258, 260
coelenterates, 35
commensalism, microbial, 126
conidia, 49
copepods, 36, 37
Corynebacterium, characteristics of, 179
cultural characteristics, bacteria, 157–60
Cyanobacteria, 30, 34, 207
- Deinococcaceae, 257
denitrification, 207, 213, 217
Deuteromycotina, 50
diatomite, 30
diatoms, 30, 33
differential WBC count, 288
disinfectants
 alcohol effectiveness, 141
 evaluation, 139
DNase test, 261

Index

- domains, classification, 25
Dorner method staining, 68
Durham tube usage, 164
- endoenzymes, 161
endospores, 67, 156
Enterobacter, 270
Enterobacteriaceae identification, 185, 189
eosinophilia, 289
eosinophils, 288, 289
epidemic, 254
erythrocytes, 288
Escherichia, 270
Eudorina, 28, 29
Euglena, 29
euglenoids, 28
Euglenophycophyta, 28
Eukarya, Domain, 25
exoenzymes, 161
- fat hydrolysis test, 172
fermentation, 161
flagellum, 28
flatworms, 35
Flavobacterium, characteristics of, 180
fluorescence method staining, 70
food spoilage, 237
formic hydrogenylase, 164
fungi, 48–53
fungi imperfecti, 50
- GasPak anaerobic jar, usage of, 90
gastrotrichs, 35
Gonium, 28, 29
gram staining, 64
green sulfur bacteria, 106
Gymnodinium, 30
- Halobacterium*, characteristics of, 180
halophile, 135
hand scrubbing evaluation, 148
hemacytometer usage, 293
hemolysis types, streptococci, 264
Henrici slide culture technique, 99
heterocysts, 207
heterotrophs, 76
hippurate hydrolysis, 268
Hirodinea, 36
hydrogen ion needs, bacterial, 77
hydrogen sulfide test, 174
hydrolysis tests for bacteria, 170
hyphae, 48
- Identibacter Interactus*, 181, 459–72
IMViC tests, 175
indole test, 190, 196
- Kirby-Bauer table, 432
Kirby-Bauer test for antibiotics, 145
Klebsiella, 270
Kovacs' reagent, usage, 173, 273
Kurthia, characteristics of, 179
- Lactobacillus brevis*, 243
characteristics of, 178
leucosin, 30
leukocytosis, 289
leukopenia, 289
Listeria, characteristics of, 178
litmus milk reactions, 176
logarithm tables, 426
lymphocytes, 288, 289
lysis, phage, 112
lysogeny, 112
- Mastigophora, 28
media preparation, 76–81
media usage
blood agar, 263, 276
Brewer's anaerobic agar, 89
desoxycholate lactose agar, 276
fluid thyoglycollate medium, 89, 159
glucose broth, 162
Hektoen enteric agar, 272
Kligler's iron agar, 174
litmus milk, 176
MacConkey agar, 272
MR-VP medium, 162
nitrate broth, 162
nutrient agar, 152
nutrient broth, 158
nutrient gelatin, 151
Russell double sugar agar, 273
semisolid medium, 155
SIM medium, 273
Simmons citrate agar, 174
skim milk agar, 170
Snyder test agar, 299
spirit blue agar, 170
starch agar, 170
trypticase soy agar, 162
xylose lysine desoxycholate agar, 272
- mesophiles, 130
metabolism, 161
metachromatic granules, 62
methyl red test, 164
microaerophiles, 89
Micrococcus, characteristics of, 179
microphages, 288
microscopy
brightfield, 2–8
darkfield, 9–11
fluorescence, 17–21
measurements, 22–24
phase contrast, 11–16
mixed acid fermentation, 164
molds, 48, 51
monocytes, 288, 289
monocytosis, 289
Morganella, 270
MPN calculation, 225
MPN table, 431
Mucor, 50, 52, 53
mycelium, 48
Myceteeae, Kingdom, 48
mycology, 48
- negative staining, 56
Neisseria, characteristics of, 181

Index

- Nematoda, 35
neutropenia, 289
neutrophilia, 289
neutrophils, 288
nitrate reduction test, 168
Nitrobacter, 206
Nitrococcus, 206
nitrogen cycle, 206
nitrogen fixation, 208
Nitrosococcus, 206
Nitrosomonas, 206
- oil immersion techniques, 7
oligodynamic action, 136
Oospora, 52, 53
optochin susceptibility test, 269
Oscillatoria, 34
osmophile, 135
osmotic pressure and growth, 135
ostracods, 36, 37
oxidase test, 168
Oxi/Ferm tube II system, 194
- palisade arrangement, 62
pandemic, 254
Pandorina, 28, 29
Paracoccus denitrificans, 218
paramylum, 28
parfocal lenses, 7
Penicillium, 50, 51
Peridinium, 30, 31
perithecia, 50
ph
 adjustment methods, 79
 effect on bacterial growth, 134
 indicators, table of, 433
Phaeophycophyta, 30
phage typing, 287
phagocytic theory of immunity, 288
phenylalanine deamination test, 175
phialospores, 49
photoautotrophs, 77
phycobilisomes, 30
phycocyanin, 30
phycoerythrin, 30
pipette handling technique, 93
Planococcus, characteristics of, 180
Plantae, 28
Plasmodium, 28
Platyhelminthes, 35
pleomorphism, 62
polychaetes, 36
population counts, bacterial
 food, 236
 meat, 239
 milk, 230
 soil, 202
population count technique, 93
pour plate techniques, 86
prokaryotes, 25, 30
Propionibacterium, characteristics of, 179
Proteus, 270
Protista, Kingdom, 26
Protozoa, Subkingdom, 26
Providencia, 270
- pseudohypha, 48
Pseudomonas, characteristics of, 180, 270
pseudopod, 28
psychrophiles, 130
purple sulfur bacteria, 106
Pyrophytophyta, 30
- reductase test, 234
resolution, microscope, 4
Rh blood typing, 298
Rhizobium, 207, 211
Rhizopus, 50, 53
rotifers, 35, 37
roundworms, 35
- Saccharomyces cerevisiae*, 241
Saccharomyces delbrueckii, 243
Salmonella, 270
salt tolerance, streptococci, 268
Sarcodina, 28
Schaeffer-Fulton method, 67
serological typing, 279
serotypes, 270
Shigella, 270
simple staining, 62
slide culture of molds, 103
slime mold culture, 100
smear preparation, 58
Smith fermentation tube, 164
soil microbiology, 201–20
solutions
 hypertonic, 135
 hypotonic, 135
 isotonic, 135
SPC, milk, 230
spectrophotometer usage, 97
spore staining, 67
Sporolactobacillus, characteristics of, 178
Sporosarcina, characteristics of, 179
Sporozoa, 28
staining
 acid-fast, 62, 69, 70
 capsular, 63
 fluorescent staining, 70
 gram, 64
 negative, 56
 simple, 62
 spore, 67
Staphylococcus, characteristics of, 180
Staphylococcus aureus, 257, 258
Staphylococcus epidermidis, 258
Staphylococcus saprophyticus, 198, 257, 258
starch hydrolysis test, 170
stigma, 28
stock cultures, 152
streak plate techniques, 82
Streptococcus, 257, 267
Streptococcus agalactiae, 262, 267
Streptococcus bovis, 262, 267
Streptococcus faecalis, 262, 267
Streptococcus pneumoniae, 262, 267
Streptococcus pyogenes, 262, 267
Submastigophora, 28
SXT sensitivity test, 267
synergism, microbial, 127

Index

- Talaromycetes*, 50
Tardigrada, 36
temperature
 effect on growth, 130
 lethal effects, 132
temperature conversion table, 488
thermal death point (TDP), 132
thermal death time (TDT), 132
thermophiles, 130
thylakoids, 30
titer, 285
Tribonema, 28, 31
trichinosis, 289
tryptophan hydrolysis test, 173
turbidimetry, 96
- ultraviolet light, lethal effects, 137
urea hydrolysis test, 173
urease, 173
urinary tract pathogens, 274
use dilution method, 139
- Vaucheria*, 28, 31
Veillonella, characteristics of, 181
Voges-Proskauer test, 192
- water bears, 36, 37
water fleas, 36, 37
Winogradsky's column, 107
Wright's stain, 290
- xanthophylls, 30
- yeasts, 48
yogurt production, 243
- Zernike microscope, 12
Ziehl-Neelsen staining method, 69
zooflagellates, 28
Zygomycotina, 50
zygospores, 49